AUTOMATED FRAUD DETECTION IN DIGITAL PAYMENTS

Industrial Internship Project report submitted in partial fulfillment of the

requirements for the Award of the Degree of

BACHELOR OF TECHNOLOGY

IN

CSE- (Artificial Intelligence & Machine Learning)

Submitted by

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(AUTONOMOUS)

Approved by (AICTE, New Delhi; Permanently Affiliated to JNTU Kakinada)

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DASARIPALEM, CHOWDAVARAM, GUNTUR - 522019 (A.P)

2025-2026

KALLAM HARANADHAREDDY INSTITUTE OF TECHNOLOGY

(AUTONOMOUS)

BACHELOR OF TECHNOLOGY

IN

CSE- (Artificial Intelligence & Machine Learning)

CERTIFICATE

This is to certify that the Industrial Internship Project report entitled **AUTOMATED FRAUD DETECTION IN DIGITAL PAYMENTS** submitted by **NARASIMHULA ANANTHA (228X1A4289)** in partial fulfillment for the award of the Degree of Bachelor of Technology in CSE-(Artificial Intelligence & Machine Learning) in the Kallam Haranadhareddy Institute of Technology and this bonafide work carried out by them during the year 2025-2026.

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Certificate ID: EXT-APSCHE_AIML-90739

Date: 20-Mar-2026

Place: Virtual

Amarendar Katkam

Founder & CEO

DECLARATION

We hereby declare that the Industrial Internship project dissertation entitled **AUTOMATED FRAUD DETECTION IN DIGITAL PAYMENTS** submitted for the B.Tech. degree is our original work and the dissertation has not formed the basis for the award of any degree, associateship, fellow- ship, or any other similar titles.

Place: Guntur
Date:

NARASIMHULA ANANTHA (228X1A4289)

ACKNOWLEDGEMENT

I am profoundly grateful to express my deep sense of gratitude and respect towards **Sri. KALLAM MOHAN REDDY, Chairman,** KHIT, Guntur for his precious support in the college.

I am thankful to **Dr. M. UMA SANKAR REDDY, M.sc, Ph.D. Secretary,** KHIT, Guntur for his Encouragement and support for the completion of the project.

I am grateful to **Mr. BHARADWAJ KALLAM, B Tech, Executive Director**, KHIT, Guntur for his Encouragement and support for the completion of the project.

I am indebted to **Mrs. NAVYA KALLAM, B.Tech, Executive Director,** KHIT, Guntur for her Encouragement and support for the completion of the project.

I am very thankful to **Dr. B. S. B REDDY, M.Tech, Ph.D, Principal,** KHIT, Guntur for his support during and until the completion of the project.

I am greatly indebted to **Dr. B. PRAKASH, M.Tech, Ph.D, Professor & Head of the Department, CSE-(Artificial Intelligence & Machine Learning),** KHIT, Guntur for providing the laboratory facilities fully as and when required and for allowing me to carry the project work in the college during the Industrial Internship.

I extend my gratitude to **Mr. E. SAMBASIVARAO, M.Tech, Project Coordinator, Associate Professor, Department of CSE-(Artificial Intelligence & Machine Learning)** for his coordination and support and for helping me in all possible ways to complete my project work on time.

I extend my deep sense of gratitude to my supervisor to **Mr. M. CHENNAKESAVA RAO, M.E, (Ph.D.), Project Guide, Assistant Professor, CSE-(Artificial Intelligence & Machine Learning)** for his guidance, and constructive ideas in every step, which were indeed of great help towards the successful completion of my project.

I express my sincere thanks to all faculty members and supporting staff for their support and for helping me in all possible ways to complete my project work on time.

Place: Guntur
Date:

By

NARASIMHULA ANANTHA (228X1A4289)

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ABSTRACT

Digital payment systems have become an integral part of modern financial ecosystems, enabling fast and convenient transactions. However, the rapid growth in digital transactions has also led to a significant rise in fraudulent activities, posing serious challenges to financial institutions. This project presents an automated fraud detection framework using the Extreme Gradient Boosting (XGBoost) algorithm to accurately identify fraudulent transactions in real time.

One of the primary challenges in fraud detection is the severe class imbalance present in financial datasets, where fraudulent transactions constitute a very small percentage of the total data. To address this issue, the Synthetic Minority Over-sampling Technique (SMOTE) is applied to balance the dataset and improve the model's ability to detect minority class instances. Additionally, domain-specific feature engineering is performed by introducing "Error Balance" metrics, which capture inconsistencies in transaction flows and enhance the model's predictive capability.

The proposed model is trained and evaluated on the PaySim dataset, achieving a high recall of 99.82% and an F1-score of 87.33%, demonstrating its effectiveness in identifying fraudulent transactions while maintaining a balance between precision and recall. Furthermore, the system is deployed using a Flask-based web application that enables real-time transaction risk classification into Low, Medium, and High categories with an average latency of 85 milliseconds.

In addition to performance, the proposed system emphasizes practical usability and scalability in real-world environments. The multi-level risk classification approach not only detects fraud but also supports decision-making by enabling appropriate actions such as transaction approval, additional verification, or blocking. This ensures minimal disruption to legitimate users while maintaining strong security controls.

The results indicate that the proposed framework provides a scalable, efficient, and practical solution for real-time fraud detection, making it suitable for integration into modern digital payment systems.

ACTIVITY LOG AND REPORT

ACTIVITY LOG FOR FIRST WEEK(FROM: 16/12/25 TO: 20/12/25)

DAY & DATE	BRIEF DESCRIPTION OF THE DAILY ACTIVITY	LEARNING OUTCOME	PERSON IN-CHARGE
TUESDAY (16/12/2025)	Introduction to Artificial Intelligence	What is AI, Uses of AI	
WEDNESDAY (17/12/2025)	Types of Artificial Intelligence	Understood about types of AI	
THURSDAY (18/12/2025)	Introduction to Machine Learning	Understood about ML	
FRIDAY (19/12/2025)	Types of Machine Learning	Understood about types of ML	
SATURDAY (20/12/2025)	Introduction to deep learning	Understood about deep learning	

WEEKLY REPORT

WEEK – 1 (16-12-2025 to 20-12-2025)

Introduction to Artificial Intelligence:

Artificial intelligence is the science of making machines that can think like humans, The goal for AI is to be able to do things such as recognize patterns, make decisions, and judge like humans.

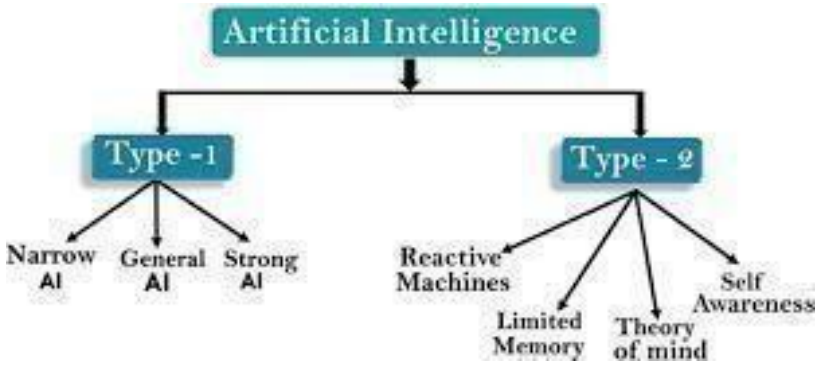
Uses of AI:

AI has been used in various applications in our daily life:

1. Digital Assistants.
2. Music and media streaming services.
3. Security.
4. Social Media.

Types of AI:

There are a lot of ongoing AI discoveries and developments, most of which are divided into different types. AI is broadly in divided in to two categories.



1) Based on capabilities:

a. Weak AI or Narrow AI: AI designed to complete very specific actions; unable to independently learn.

b. General AI: AI designed to learn, think and perform at similar levels to humans.

c. Super AI: AI able to surpass the knowledge and capabilities of humans.

2) Based on functionality:

a. Reactive Machines: AI capable of responding to external stimuli in real time; unable to build memory or store information for future.

b. Limited memory machines: AI that can store knowledge and use it to learn and train for future tasks.

c. Theory of Mind: AI that can sense and respond to human emotions, plus perform the tasks of limited memory machines.

d. Self-awareness: AI that can recognize others' emotions, plus has sense of self and human-level intelligence; the final stage of AI.

Introduction to Machine Learning:

Machine learning (ML) is a branch of artificial intelligence (AI) that enables computers to "self-learn" from training data and improve over time, without being explicitly programmed. Machine learning algorithms are able to detect patterns in data and learn from them, in order to make their own predictions.

Uses of ML:

ML is widely used in following applications:

- **E -commerce Product recommendations:** It detect items that match the customer's needs based on the customer's previous actions and make them appear in an interesting way.
- **Self-Driving Cars:** Machine learning allow a car to collect data on its surroundings from cameras and other sensors, interpret it, and decide what actions to take.
- **Email automation and spam filtering:** It identify and classify incoming spam emails based on their resemblance to stored training examples of spam emails.

- **Chatbots:** Chatbots are used a lot in customer interaction, marketing on social network sites, and instant messaging the client.

Types of ML: Based on the methods and way of learning, machine learning is divided into mainly four types, which are:

Supervised machine learning: Supervised learning is a category of machine learning that uses labeled datasets to train algorithms to predict outcomes and recognize patterns.

1. **Unsupervised machine learning:** It is a type of learning where models are given unlabeled data and allowed to discover patterns and insights without any explicit guidance or instruction.
2. **Semi-Supervised learning:** Semi-supervised learning is a branch of machine learning that combines supervised and unsupervised learning by using both labeled and unlabeled data and it is a a broad category of machine learning that uses labeled data to ground predictions, and unlabeled data to learn the shape of the larger data distribution.
3. **Reinforcement machine learning:** Reinforcement learning (RL) is a machine learning (ML) technique that trains software to make decisions to achieve the most optimal results.

It mimics the trial-and-error learning process that humans use to achieve their goals.

Introduction to deep Learning: Deep learning is a branch of machine learning which is based on artificial neural networks. It is capable of learning complex patterns and relationships within data. In deep learning, we don't need to explicitly program. Everything. It has become increasingly popular in recent years due to the advances in processing power and the availability of large datasets. Because it is based on artificial neural networks (ANNs) also known as deep neural networks (DNNs). These neural networks are inspired by the structure and function of the human brain's biological neurons, and they are designed to learn from large amounts of data.

ACTIVITY LOG FOR SECOND WEEK(FROM: 22/12/25 TO: 24/12/25)

DAY & DATE	BRIEF DESCRIPTION OF THE DAILY ACTIVITY	LEARNING OUTCOME	PERSON IN-CHARGE
MONDAY (22/12/2025)	Python Basics & Packages	Understood about Python Basics and Packages.	
TUESDAY (23/12/2025)	Data Visualization	Understood about Data Visualization	
WEDNESDAY (24/12/2025)	EDA with Matplotlib & Seaborn	Understood about Matplotlib & Seaborn	

WEEKLY REPORT

WEEK – 2 (22-12-2025 to 24-12-2025)

Python Basics: Data science is an interconnected field that involves the use of statistical and computational methods to extract insightful information and knowledge from data. Python is a popular and versatile programming language, now has become a popular choice among data scientists for its ease of use, extensive libraries, and flexibility. Python provide and efficient and streamlined approach to handing complex data structure and extracts insights.

Python Packages: Python is a computer programming language often used to build websites and software, automate tasks, and analyze data.

Python library is a collection of related modules. It contains bundles of code that can be used repeatedly in different programs. It makes Python Programming simpler and convenient for the programmer.

Pandas: Pandas is a Python library used for working with data sets. It has functions for analyzing, cleaning, exploring, and manipulating data.

Matplotlib: Matplotlib is a cross-platform, data visualization and graphical plotting library (histograms, scatter plots, bar charts, etc) for Python and its numerical extension NumPy. As such, it offers a viable open source alternative to MATLAB.

Numpy: NumPy is a very popular python library for large multi-dimensional array and matrix processing, with the help of a large collection of high-level mathematical functions.

Tensorflow: TensorFlow is a very popular open-source library for high performance numerical computation developed by the Google Brain team in Google

Data Visualization:

Data visualization is the practice of translating information into a visual context, such as a map or graph, to make data easier for the human brain to understand and pull insights from. The main goal of data visualization is to make it easier to identify patterns, trends and outliers in large data sets.

Techniques: It involves using various visual elements such as charts, graphs, maps, and infographics to communicate insights effectively.

Bar Charts: A bar chart or bar graph is a chart that represents categorical data with rectangular bars with heights proportional to the values that they represent.

Line Charts: A line chart is generally constructed by marking points that correspond to each data value ordered according to a criterion and then joining these by lines

Pie Charts: It displays the pictorial representation of data that makes it possible to visualize the relationships between the parts and the whole of a variable.

Scatter Plots: Scatter plots are the graphs that present the relationship between two variables in a data-set.

Heatmaps: Represent data values in a matrix format using colors to indicate the magnitude of the values.

Matplotlib:

It's fast, efficient as it is based on numpy and also easier to build.

Has undergone a lot of improvements from the open source community since inception and hence a better library having advanced features as well.

Well maintained visualization output with high quality graphics draws a lot of users to it Basic as well as advanced charts could be very easily built.

Seaborn:

- Built-in themes aid better visualization.
- Statistical functions aiding better data insights.
- Better aesthetics and built-in plots.
- Helpful documentation with effective examples.

ACTIVITY LOG FOR THIRD WEEK(FROM: 26/12/25 TO: 31/12/25)

DAY & DATE	BRIEF DESCRIPTION OF THE DAILY ACTIVITY	LEARNING OUTCOME	PERSON IN-CHARGE
FRIDAY (26/12/2025)	Introduction to Preprocessing	What is Preprocessing	
SATURDAY (27/12/2025)	Handing Missing Data	Understood about Handling Missing Data	
MONDAY (29/12/2025)	Handling Outliers Data	Understood about Handling Outliers Data	
TUESDAY (30/12/2025)	Splitting the data into Train & Test set	Splitting the data into Train & Test set	
WEDNESDAY (31/12/2025)	Feature Selection	Understood about Feature Selection	

WEEKLY REPORT

WEEK – 3 (26-12-2025 to 31-12-2025)

Introduction to Preprocessing: Data preprocessing is the process of generating raw data for machine learning models. This is the first step in creating a machine learning model. This is the most complex and time consuming aspect of data science. Data preprocessing is required in machine learning algorithms to reduce its complexities. Steps in Preprocessing: There are six steps of data preprocessing in machine learning

- Import the Libraries.
- Import the Loaded Data.
- Check for Missing Values.
- Arrange the Data.
- Do Scaling.
- Distribute Data into Training, Evaluation and Validation Sets.

Handling Missing Data: Missing values are data points that are absent for a specific variable in a dataset. They can be represented in various ways, such as blank cells, null values, or special symbols like “NA” or “unknown.” These missing data points pose a significant challenge in data analysis and can lead to inaccurate or biased results.

Handling Outliers Data: Outliers can be caused by measurement uncertainty or due to experimental error. Outliers in data can be observed using a number of techniques. To find outliers, we can simply plot the box plot. Outliers are points that are outside of the minimum and maximum values. Outlier detection and handling are crucial aspects of building reliable and robust machine learning models. Understanding the impact of outliers, choosing the appropriate technique for your specific data and task, and leveraging domain knowledge and data visualization can ensure that your models perform well on unseen data and provide accurate and trustworthy predictions.

Train & Test Set: Train and test datasets are the two key concepts of machine learning, where the training dataset is used to fit the model, and the test dataset is used to evaluate the model. we can easily evaluate the performance of our model. Such as, if it performs well with the training data, but does not perform well with the test dataset, then it is estimated that the model may be overfitted.

For splitting the dataset, we can use the **train_test_split** function of **scikit-learn.**

Feature selection: Feature selection is one of the important concepts of machine learning, which highly impacts the performance of the model. As machine learning works on the concept of "Garbage In Garbage Out", so we always need to input the most appropriate and relevant dataset to the model in order to get a better result.

○ It helps in avoiding the curse of dimensionality. ○ It helps in the simplification of the model so that it can be easily interpreted by the researchers. ○ It reduces the training time.
It reduces overfitting hence enhance the generalization

ACTIVITY LOG FOR FOURTH WEEK(FROM: 02/01/26 TO: 06/01/26)

DAY & DATE	BRIEF DESCRIPTION OF THE DAILY ACTIVITY	LEARNING OUTCOME	PERSON IN-CHARGE
FRIDAY (02/01/2026)	Introduction to Regression	Understood about Regression concepts	
SATURDAY (03/01/2026)	Types of Regression	Understood about Linear & Multi Regression models	
MONDAY (05/01/2026)	Introduction to Logistic Regression	Understood about Logistic Regression	
TUESDAY (06/01/2026)	Types of Logistic Regression	Understood about types Logistic Regression.	

WEEKLY REPORT

WEEK – 4 (02-01-2026 to 06-01-2026)

Introduction to Regression: Regression is a supervised machine learning technique which is used to predict continuous values. The ultimate goal of the regression algorithm is to plot a best-fit line or a curve between the data.

Types of Regresssion:

Mainly there are two types of regression. They are:

- **Simple Linear Regression:** It is a type of Regression algorithms that models the relationship between a dependent variable and a single independent variable. The relationship shown by a Simple Linear Regression model is linear or a sloped straight line, hence it is called Simple Linear Regression.

- **Multi Linear Regression:** Multiple Linear Regression is an extension of Simple Linear regression as it takes more than one predictor variable to predict the response variable.

Logistic Regression: Logistic regression is used for binary classification where we use sigmoid function, that takes input as independent variables and produces a probability value between 0 and 1. Logistic regression predicts the output of a categorical dependent variable. Therefore, the outcome must be a categorical or discrete value.

Types of Logistic Regression:

On the basis of the categories, Logistic Regression can be classified into three types:

Binomial: In binomial Logistic regression, there can be only two possible types of the dependent variables, such as 0 or 1, Pass or Fail, etc.

Multinomial: In multinomial Logistic regression, there can be 3 or more possible unordered types of the dependent variable, such as “cat”, “dogs”, or “sheep”

Ordinal: In ordinal Logistic regression, there can be 3 or more possible ordered types of dependent variables, such as “low”, “Medium”, or “High”.

ACTIVITY LOG FOR FIFTH WEEK(FROM: 07/01/26 TO: 10/01/26)

DAY & DATE	BRIEF DESCRIPTION OF THE DAILY ACTIVITY	LEARNING OUTCOME	PERSON IN-CHARGE
WEDNESDAY (07/01/2026)	Introduction to Decision trees	Understood about Decision trees	
THURSDAY (08/01/2026)	Decision Trees Continution	Understood about terminologies and types of decision trees	
FRIDAY (09/01/2026)	Introduction to Random forest	Understood about random forest algorithm	
SATURDAY (10/01/2026)	KNN Algorithm	Understood about KNN algorithm	

WEEKLY REPORT

WEEK – 5 (07-01-2026 to 10-01-2026)

Decision Trees: In machine learning, a decision tree is a predictive model that uses a splits the data into 10 subsets based on features, with each split maximizing the homogeneity of the target variable within the subsets. Decision trees are used for classification and regression tasks, offering interpretability and versatility in handling various types of data.

Terminologies in Decision Tree:

- Root Node
- Decision Nodes
- Leaf Nodes
- Sub-Tree
- Pruning
- Sub-Tree
- Parent and Child Node

Steps in Decision tree:

- Starting at the Root
- Asking the Best Questions
- Branching Out
- Repeating the Process

Random forest: It is also called a lazy learner algorithm because it does not learn from the training set immediately instead it stores the dataset and at the time of classification, it performs an action on the dataset.

Uses of Random Forest:

1. It takes less training time as compared to other algorithms.
2. It predicts output with high accuracy, even for the large dataset it runs efficiently.
3. It can also maintain accuracy when a large proportion of data is missing.

The Working process can be explained in the below steps:

Step-1: Select random K data points from the training set.

Step-2: Build the decision trees associated with the selected data points (Subsets).

Step-3: Choose the number N for decision trees that you want to build.

Step-4: Repeat Step 1 & 2.

Step-5: For new data points, find the predictions of each decision tree, and assign the new data points to the category that wins the majority votes.

KNN algorithm: It is also called a lazy learner algorithm because it does not learn from the training set immediately instead it stores the dataset and at the time of classification, it performs an action on the dataset.

The KNN working can be explained on the basis of the below algorithm:

Step-1: Select the number K of the neighbors

Step-2: Calculate the Euclidean distance of K number of neighbors

Step-3: Take the K nearest neighbors as per the calculated Euclidean distance.

Step-4: Among these k neighbors, count the number of the data points in each category.

Step-5: Assign the new data points to that category for which the number of the neighbor is maximum.

Step-6: Our model is ready.

ACTIVITY LOG FOR SIXTH WEEK(FROM: 12/01/26 TO: 17/01/26)

DAY & DATE	BRIEF DESCRIPTION OF THE DAILY ACTIVITY	LEARNING OUTCOME	PERSON IN-CHARGE
THURSDAY (12/01/2026)	Introduction to Bayes Theorem And Classification	Understood about Bayes Theorem and Classification	
FRIDAY (13/01/2026)	Types of classification	Linear & NonLinear Models	
WEDNESDAY (17/01/2026)	Linear Models& Non Linear models	Understood about Linear Models and Non Linear Models	

WEEKLY REPORT

WEEK – 6 (12-01-2026 to 17-01-2026)

Bayes Theorem: Bayes theorem is one of the most popular machine learning concepts that helps to calculate the probability of occurring one event with uncertain knowledge while other one has already occurred.

Bayes' theorem can be derived using product rule and conditional probability of event X with known event Y.

The theorem can be mathematically expressed as: P(A|B) = P(B|A)P(A) / P(B) Where,

- P(A|B) is the posterior probability of event A given event B.
- P(B|A) is the likelihood of event B given event A.
- P(A) is the prior probability of event A.
- P(B) is the total probability of event B.

Classification: Classification is a supervised machine learning method where the model tries to predict the correct label of a given input data. The Classification algorithm is a Supervised Learning technique that is used to identify the category of new observations on the basis of training data. In Classification, a program learns from the given dataset or observations and then classifies new observation into a number of classes or groups. Such as, **Yes or No, 0 or 1, Spam or Not Spam, cat or dog,** etc. Classes can be called as targets/labels or categories.

Unlike regression, the output variable of Classification is a category, not a value, such as "Green or Blue", "fruit or animal", etc. Since the Classification algorithm is a Supervised learning technique, hence it takes labeled input data, which means it contains input with the corresponding output.

Types of Classification: Classification Algorithms can be further divided into the mainly two category:
Linear Models:

1. Logistic Regression
2. Support Vector Machines

Non Linear Models:

a. K-Nearest Neighbors

b. Kernel SVM

c. Decision Tree Classification

ACTIVITY LOG FOR SEVENTH WEEK(FROM: 19/01/26 TO: 24/01/26)

DAY & DATE	BRIEF DESCRIPTION OF THE DAILY ACTIVITY	LEARNING OUTCOME	PERSON IN-CHARGE
MONDAY (19/01/2026)	Introduction to Ensemble Samplings	Understood about Ensemble Samplings	
TUESDAY (20/01/2026)	Ensemble Sampling Techniques	Understood about Ensemble Samplings techniques	
WEDNESDAY (21/01/2026)	Support Vector Machine	Understood about Support Vector Machine	
THURSDAY (22/01/2026)	Types of Support Vector Machine	Working understood about types of SVM	
FRIDAY (23/01/2026)	Hyper Parameter Tuning	Understand about Hyper Parameter tuning	
SATURDAY (24/01/2026)	Introduction to Ensemble Samplings	Understood about Ensemble Samplings	

WEEKLY REPORT

WEEK –7 (19-01-2026 to 24-01-2026)

Ensembling Sampling: Ensemble learning is an approach in which two or more models are fitted to the same data, and the predictions of each model are combined. Ensemble learning aims to achieve better performance with the ensemble of models than with any individual model. There are several methods of ensembling techniques in machine learning, including:

1. **Voting:** Different models make predictions, and the final prediction is determined by a majority vote or averaging their predictions.
2. **Bagging (Bootstrap Aggregating):** Multiple models (often of the same type) are trained on different subsets of the training data, typically with replacement. The final prediction is an average or voting of these models.
3. **Boosting:** Models are trained sequentially, where each subsequent model corrects the errors of the previous one. Examples include AdaBoost, Gradient Boosting Machines (GBM), and XGBoost.
4. **Stacking:** Predictions from multiple models are used as input features to a meta model, which then produces the final prediction. It combines the strengths of different models.
5. **Random Forest**: An ensemble method based on decision trees, where multiple decision trees are trained on different subsets of the data, and the final prediction is determined by averaging or voting. These methods help improve the overall performance and robustness of machine learning models by leveraging the diversity of multiple models.

Support Vector Machine: Support Vector Machine, abbreviated as SVM can be used for both regression and classification tasks, but generally, they work best in classification problems. The main objective of the SVM algorithm is to find the optimal hyper plane in an N-dimensional space that can separate the data points in different classes in the feature space. The hyper plane tries that the margin between the closest points of different classes should be as maximum as possible. **Types of Support Vector machine:**

Support Vector Machines (SVM) can be divided into two main parts:

Linear SVM: Typically used for linear regression and classification problems. Linear SVM is used for linearly separable data, which means if a dataset can be classified into two classes by using a

single straight line, then such data is termed as linearly separable data, and classifier is used called as Linear SVM classifier.

Non Linear SVM: Non-Linear SVM is used for non-linearly separated data, which means if a dataset cannot be classified by using a straight line, then such data is termed as non-linear data and classifier used is called as Nonlinear SVM classifier.

Hyperparameter Tuning: Hyperparameters directly control model structure, function, and performance. Hyperparameter tuning allows data scientists to tweak model performance for optimal results. This process is an essential part of machine learning, and choosing appropriate hyperparameter values is crucial for success. The goal of hyperparameter tuning is to find the values that lead to the best performance on a given task.

Types of Hyperparameter tuning:

- **Number of hidden layers:** It's a trade-off between keeping our neural network as simple as possible (fast and generalized) and classifying our input data correctly.
- **Number of nodes/neurons per layer:** More isn't always better when determining how many neurons to use per layer. Increasing neuron count can help, up to a point.
- **Learning rate:** Model parameters are adjusted iteratively and the learning rate controls the size of the adjustment at each step.
- **Momentum:** Momentum helps us avoid falling into local minima by resisting rapid changes to parameter values.

ACTIVITY LOG FOR EIGHTH WEEK(FROM: 27/01/26 TO: 31/01/26)

DAY & DATE	BRIEF DESCRIPTION OF THE DAILY ACTIVITY	LEARNING OUTCOME	PERSON IN-CHARGE
TUESDAY (27/01/2026)	Introduction to Flask	Flask and its uses	
WEDNESDAY (28/01/2026)	Flask Framework Process	Setting up Flask frame work	
THURSDAY (29/01/2026)	Flask templating HTML files	Setting up Flask and running	
FRIDAY (30/01/2026)	Deploying ML Models on Flask	Understood how to Deploy ML Models on Flask	
SATURDAY (31/01/2026)	Flask Functionalities	Understood importance of Flask Functionalities	

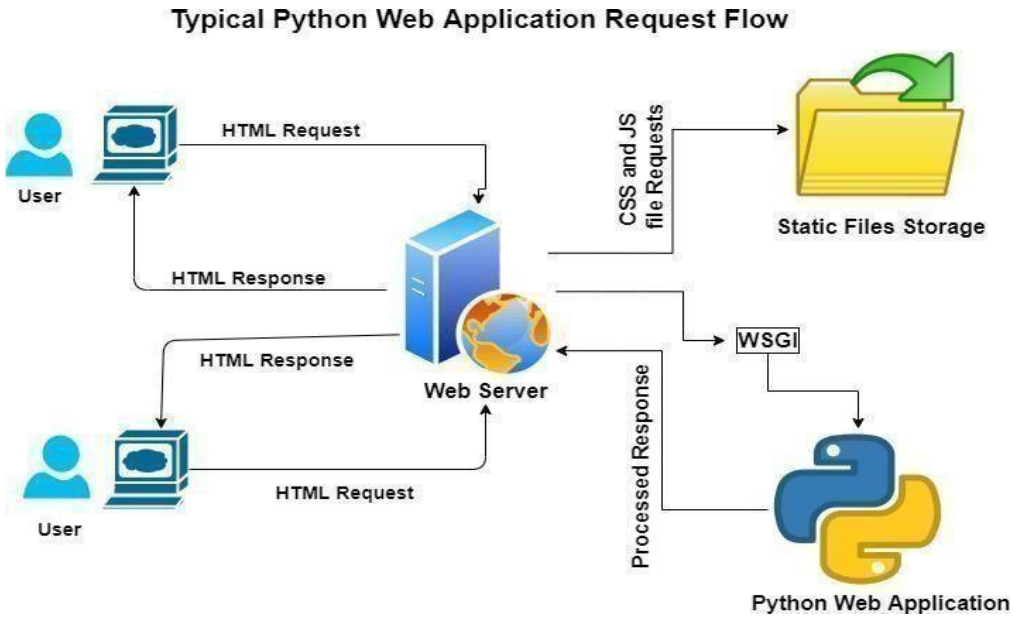
WEEKLY REPORT

WEEK –8 (27-01-2026 to 31-01-2026)

Introduction to Flask:

1) Flask is a Micro Web framework written in python.
2) Flask is based on WSGI, Template Engine.

Flask Framework Process:



Flask Templating in HTML Files and Deployment: Steps:

1) Setting up Environment for creating flask application.
2) Create an HTML file for rendering.
3) Create an app.py file and place it outside of HTML file.
4) Run the app.py file.

Deploying ML Models on Flask:

1. Train and Save the Model
2. Create a Flask App

3. Set Up Flask App Structure
4. Load the Model
5. Create Flask Routes
6. Create HTML Templates
7. Run the Flask App Locally
8. Deploy to a Web Server

Routing and URL mapping: Flask allows developers to map different URLs to specific functions, making it easy to handle different types of requests and build API endpoints for data retrieval.

Templating: Flask's templating engine enables dynamic HTML generation, allowing data scientists to integrate their analyses and visualizations seamlessly into web pages.

Integration with data science libraries: Flask seamlessly integrates with popular data science libraries like NumPy, Pandas, and Matplotlib, enabling data scientists to leverage their powerful functionality in web applications.

Form handling and user inputs: Flask simplifies the process of handling form submissions, making it straightforward to collect user input for further analysis or processing.

Database integration: Flask can easily connect to databases through extensions like Flask SQL, facilitating the storage and retrieval of data within web applications.

ACTIVITY LOG FOR NINETH WEEK(FROM: 02/02/26 TO: 06/02/26)

ACTIVITY LOG FOR NINETH WEEK(FROM: 02/02/26 TO: 06/02/26)

DAY & DATE	BRIEF DESCRIPTION OF THE DAILY ACTIVITY	LEARNING OUTCOME	PERSON IN-CHARGE
MONDAY (02/02/2026)	Introduction to Artificial Neural Network	Working and understood about types and perceptron	
TUESDAY (03/02/2026)	ANN Architecture	Understood about ANN	
WEDNESDAY (04/02/2026)	Working of ANN	Hands on implementation	
THURSDAY (05/02/2026)	ANN Activation Function	Understood about ANN Activation Function	
FRIDAY (06/02/2026)	Types of ANN	Understood about Forward &Backword Propagation	

WEEKLY REPORT

WEEK –9 (02-02-2026 to 06-02-2026)

Introduction to ANN: An Artificial Neural Network (ANN) was developed by Jerfery Hiltn. ANN is similar to human brain's neural structure. It consists of inter connected nodes (neurons) organized into layers. Information flows through these node, and the network adjusts the connection strengths (weights) during training to learn from data. Enabling it to recognize patterns, make predictions, and solve various tasks in machine learning and artificial intelligence.

Artificial Neural Networks Architecture: The term "Artificial Neural Network" is derived from Biological neural networks that develop the structure of a human brain. These neurons are known as nodes.

ANN contains mainly three layers:

1) Input Layer :- Accepts all inputs that are given by the user.
2) Hidden Layer :- Performs all the calculations to find hidden features and patters.
3) Output Layer :- Predicts the Output based on input and weighted calculation at hidden neurons

ANN Activation function: This summed function is applied over an Activation function. The output from this neuron is multiplied with the weight W3 and supplied as input to the output layer. (Y=W1X1+W2X2+b)

Types of ANN:

Feed Forward Neural Network: Feedforward neural network is a linear network, which consists of a single layer of output nodes, the inputs are fed directly to the outputs via a series of weights. The sum of the products of the weights and the inputs is calculated in each node

Feed Backward propagation: Backward propagation is an algorithm that is designed to test for errors working back from output nodes to input nodes. If there are any errors then by using this algorithm we can prevent the rate of errors

ACTIVITY LOG FOR TENTH WEEK(FROM: 09/02/26 TO: 13/02/26)

DAY & DATE	BRIEF DESCRIPTION OF THE DAILY ACTIVITY	LEARNING OUTCOME	PERSON IN-CHARGE
MONDAY (09/02/2026)	Introduction to CNN	CNN and Uses	
TUESDAY (10/02/2026)	Architecture of CNN	Understood how CNN works	
WEDNESDAY (11/02/2026)	Introduction to Transfer Learning	Understood about Transfer Learning	
THURSDAY (12/02/2026)	Types of Transfer Learning	Different Types of Transfer Learning	
FRIDAY (13/02/2026)	Use Cases of Machine Learning	Learned multiple Use cases	

WEEKLY REPORT

WEEK –10 (09-02-2026 to 13-02-2026)

Introduction to CNN:

Convolutional neural network (CNN) is a network architecture for deep learning that learns directly from data. CNNs are particularly useful for finding patterns in images to recognize objects, classes, and categories. A CNN is a powerful tool but requires millions of labelled data points for training.

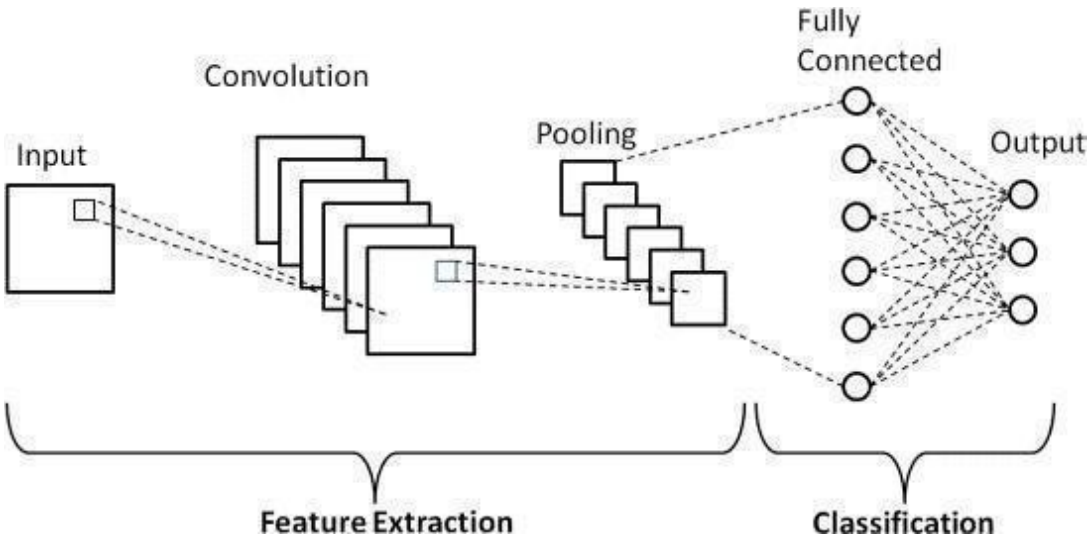
Applications of CNN:

1) **Image classification:** CNNs are the state-of-the-art models for image classification. They can be used to classify images into different categories, such as cats and dogs.
2) **Object detection:** CNN can detect and locate objects in images or videos like people, cars, and buildings. They can also be used to localize objects in images, which means that they can identify the location of an object in an image.
3) **Image segmentation:** CNNs can be used to segment images, which means that they can identify and label different objects in an image. This is useful for applications such as medical imaging.

CNN Architecture:

The CNN's job is to compress the images into a format that is easier to process while preserving elements that are important for obtaining a decent prediction. A convolutional neural network, CNN in short has three layers which are its building blocks, let's have a look:

1) Convolutional Layer.
2) Pooling layer.
3) Fully Connected Layer.



Types of layers:

Input Layers: It's the layer in which we give input to our model. In CNN, Generally, the input will be an image or a sequence of images. This layer holds the raw input of the image with width 32, height 32, and depth 3.

Convolutional Layers: This is the layer, which is used to extract the feature from the input dataset. It applies a set of learnable filters known as the kernels to the input images. The filters/kernels are smaller matrices usually 2×2, 3×3, or 5×5 shape. it slides over the input image data and computes the dot product between kernel weight and the corresponding input image patch. The output of this layer is referred as feature maps.

Pooling layer: This layer is periodically inserted in the CNN and its main function is to reduce the size of volume which makes the computation fast reduces memory and also prevents overfitting. Two common types of pooling layers are max pooling and average pooling.

Fully Connected Layer: Fully Connected Layer is also known as Dense Layer. The fully connected layer plays a critical role in the final stages of a CNN, where it is responsible for classifying images based on the features extracted in the previous layers. The term fully connected means that each neuron in one layer is connected to each neuron in the subsequent layer.

Activation Function: The activation function is typically applied to the output of each neuron in the network. It takes in the weighted sum of the inputs and produces an output.

Introduction to Transfer Learning:

Transfer learning (TL) is a technique in machine learning (ML) in which knowledge learned from a task is re-used in order to boost performance on a related task. For example, for image classification, knowledge gained while learning to recognize cars could be applied when trying to recognize trucks. There are two ways to make use of knowledge from the pretrained model. The first is to freeze some layers from the pretrained model and then train layers using our new dataset. The second way is to create a new model but also remove some features from the layer in the pre-trained model.

Types of transfer learning:

Deep transfer learning is a technique that utilizes pre-trained deep neural networks as the starting point for training on a new task. There are several different types of algorithms used in deep transfer learning, including:

1) **Fine-tuning:** This involves taking a pre-trained network and training it further on a new task by adjusting the weights of the final layers.
2) **Multi-task Learning:** Training a single model to perform multiple tasks simultaneously, where the knowledge learned from one task can benefit the performance on other related tasks.

Feature extraction: This method uses the features learned by a pretrained network as input to a new classifier, which is trained from scratch.

Use Cases of Machine Learning:

Image Recognition: Identifying objects, people, places, and activities in images and videos, used in applications like facial recognition, medical imaging, and autonomous vehicles.

Recommendation Systems: Suggesting products, services, or content based on user preferences and behavior, seen in e-commerce platforms, streaming services, and social media platforms.

ACTIVITY LOG FOR ELEVENTH WEEK(FROM: 16/02/26 TO: 20/02/26)

DAY & DATE	BRIEF DESCRIPTION OF THE DAILY ACTIVITY	LEARNING OUTCOME	PERSON IN-CHARGE
MONDAY (16/02/2026)	Model Optimization Techniques	Understood the importance of optimization and fine-tuning	
TUESDAY (17/02/2026)	Regularization Methods (L1, L2, Dropout)	Understood how regularization prevents overfitting	
WEDNESDAY (18/02/2026)	Model Evaluation Metrics	Learned about accuracy, precision, recall, F1-score	
THURSDAY (19/02/2026)	Confusion Matrix & ROC Curves	Understood performance visualization for classifiers	
FRIDAY (20/02/2026)	Hyperparameter Optimization (Grid/Random Search)	Implemented model tuning using scikit- learn	

WEEKLY REPORT

WEEK –11 (16-02-2026 to 20-02-2026)

Model optimization:

Model optimization techniques used in improving the efficiency and performance of machine learning algorithms. Explored different optimizers such as Adam, RMSprop, and SGD, and understood how learning rate scheduling can prevent underfitting or overfitting. Practical examples helped visualize how these optimizers affect convergence speed during training.

Regularization methods:

Regularization methods like L1 and L2 to prevent overfitting in machine learning models. Understood how the regularization terms add penalties to large coefficients, keeping the model generalizable. Dropout regularization was implemented in a neural network to observe its effect on model robustness and performance.

Evaluation metrics:

Evaluation metrics for classification and regression models. Learned how to calculate and interpret accuracy, precision, recall, and F1- score. Explored real-world examples to understand how relying on accuracy alone can be misleading in imbalanced datasets.

Analyzed the confusion matrix and Receiver Operating Characteristic (ROC) curve. Understood how the Area Under the Curve (AUC) measures the ability of a classifier to distinguish between classes. Generated confusion matrices for different models to evaluate prediction errors.

Hyperparameter tuning:

Hyperparameter tuning using GridSearch and Randomized SearchCV from the scikit-learn library. Understood how adjusting parameters such as depth, learning rate, and estimator count improves model performance. Compared results from different models to find optimal configurations.

ACTIVITY LOG FOR TWELEVTH WEEK(FROM: 23/02/26 TO: 27/02/26)

DAY & DATE	BRIEF DESCRIPTION OF THE DAILY ACTIVITY	LEARNING OUTCOME	PERSON IN-CHARGE
MONDAY (23/02/2026)	Introduction to Deep Learning Frameworks	Understood Tensor Flow & Keras basics	
TUESDAY (24/02/2026)	Building ANN &CNN Models using Keras	Implemented model using Sequential API	
WEDNESDAY (25/02/2026)	Model Compilation and Training	Learned optimizer selection and loss functions	
THURSDAY (26/02/2026)	Model Evaluation and Visualization	Evaluated accuracy, loss curves, and validation	
FRIDAY (27/02/2026)	Saving and Loading Models	Learned how to save trained models (.h5format)	

WEEKLY REPORT WEEK –12 (23-02-2026 to 27-02-2026)

Deep Learning Frameworks:

Deep learning ecosystem and learned about the TensorFlow and Keras frameworks. Understood the purpose of each library and how Keras simplifies neural network implementation with high-level APIs. Set up the development environment and successfully executed basic neural network scripts.

Building ANN & CNN:

Built Artificial Neural Network (ANN) and Convolutional Neural Network (CNN) models using the Keras Sequential API. Configured different layers such as Dense, Conv2D, MaxPooling, and Flatten. Learned to apply activation functions like ReLU and Softmax for non-linearity and classification tasks.

Leanring About the Models:

Focused on compiling and training deep learning models. Selected optimizers like Adam and RMSprop, and experimented with loss functions such as categorical cross-entropy and mean squared error. Monitored model learning across multiple epochs.

Visualized model performance through accuracy and loss graphs. Learned how to track validation accuracy to detect overfitting or underfitting. Understood how to use callbacks such as Early Stopping to control training flow.

Saved the trained models using the .h5 format and learned how to reload them for inference.
Understood the importance of saving models for deployment and real-time prediction application

ACTIVITY LOG FOR PROJECT
(FROM: 02/03/26 TO: 31/03/26)

Automated Fraud Detection in Digital Payments

1. PROJECT DESCRIPTION

The digital payment ecosystem is a cornerstone of the modern global economy, yet it is increasingly targeted by sophisticated cyber-criminals. Fraudulent transactions not only lead to massive financial losses for businesses but also erode consumer trust in digital banking and e- commerce platforms. As fraud techniques evolve, traditional rule-based systems which rely on static "if-then" logic are no longer sufficient to catch complex, shifting patterns of criminal behavior.

The core challenge in Online Payments Fraud Detection is the ability to distinguish between a legitimate customer and a fraudster in a matter of milliseconds. By leveraging Machine Learning (ML), we can analyze thousands of transaction features simultaneously to identify high-risk activity that would be invisible to the human eye. This project focuses on moving from a reactive "dispute-based" model to a proactive, real-time prevention model.

We will implement and compare several high-performance classification algorithms, including Decision Trees, Random Forest, K-Nearest Neighbors (KNN), and XGBoost. Given the nature of fraud data, we will pay special attention to class imbalance (where fraudulent transactions are a tiny fraction of the total data) to ensure the model doesn't just "guess" that every transaction is safe.

2. PRE-REQUISITES

To complete this project, you will require the following software, concepts, and packages Anaconda navigator:

 Refer to the link below to download the anaconda navigator

3. PYTHON PACKAGES

- Open anaconda prompt as administrator
- Type “pip install numpy” and click enter.
- Type “pip install pandas” and click enter.

- Type “pip install scikit-learn” and click enter.
- Type ”pip install matplotlib” and click enter.
- Type ”pip install scipy” and click enter.
- Type ”pip install pickle-mixin” and click enter.
- Type ”pip install seaborn” and click enter.
- Type “pip install Flask” and click enter.

4. PRIOR KNOWLEDGE

- Supervised Learning
- Unsupervised Learning
- Metrics
- Flask

5. PROJECT OBJECTIVES

By the end of this project you will:

- Know fundamental concepts and techniques used for machine learning.
- Gain a broad understanding of data.
- Have knowledge of pre-processing the data/transformation techniques and some visualization concepts before building the model
- Learn how to build a machine learning model and tune it for better performance
- Know how to evaluate the model and deploy it using flask

6. PROJECT FLOW

- The user interacts with the UI to enter the input.
- Entered input is analyzed by the model which is integrated.
- The predictions made by the model are showcased on the UI

To accomplish this, we have to complete all the activities listed below,

Data collection

- Collect the dataset or create the dataset

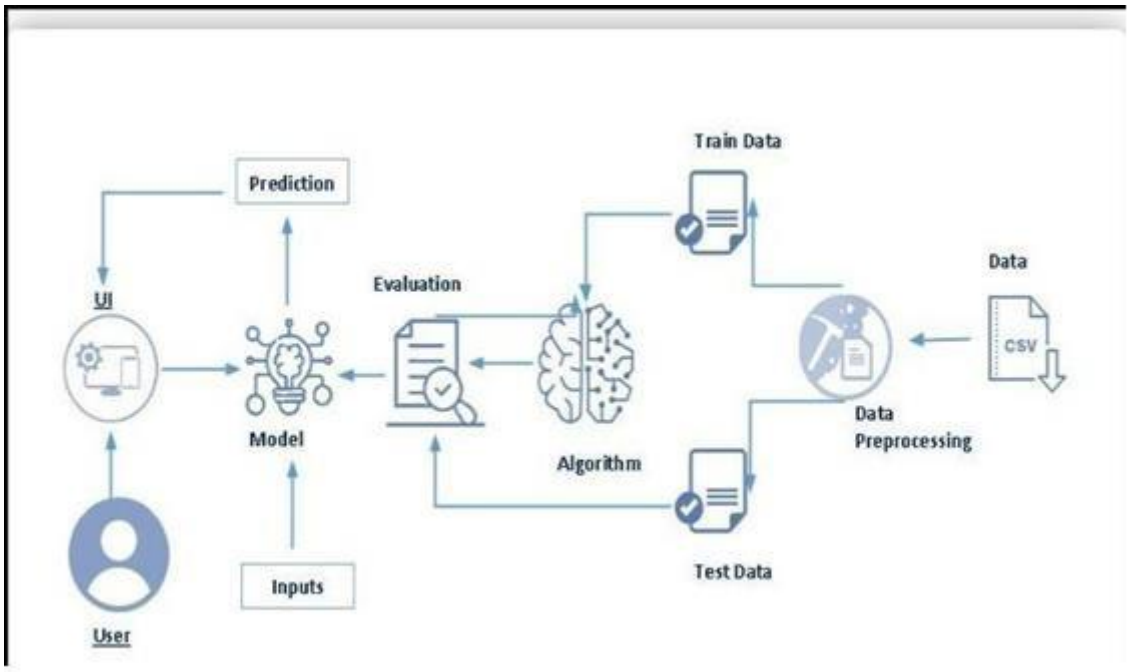
Data pre-processing

- Removing unnecessary columns
- Checking for null values
- **Visualizing and analyzing data**
- **Univariate analysis**
- **Bivariate analysis**
- **Descriptive analysis** **Model building**
- Handling categorical values
- Dividing data into train and test sets
- Import the model building libraries
- Comparing the accuracy of various models
- Hyperparameter tuning of the selected model
- Evaluating the performance of models
- Save the model

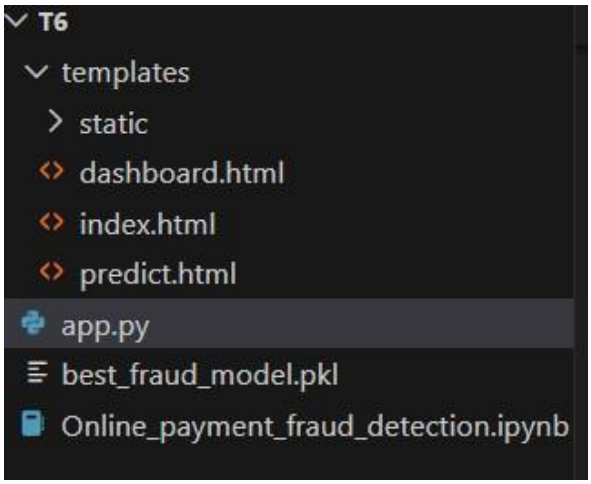
Application Building

- Create an HTML file
- Build python code

7. PROJECT ARCHITECTURE



Create the Project folder which contains files as shown below



- We are building a flask application which needs HTML pages stored in the templates folder and a python script app.py for scripting.
- Best_fraud_model is our saved model. Further we will use this model for flask integration.

7.1 Milestone 1: Data Collection

Machine Learning depends heavily on high-quality data; it is the most crucial aspect that makes algorithm training and pattern recognition possible. In the context of financial security, the dataset must capture a wide variety of transaction types to allow the model to distinguish between "Normal" and "Fraudulent" behavior.

Activity 1: Data Acquisition

There are many popular open-source repositories for collecting financial and behavioral data, such as Kaggle and the UCI Machine Learning Repository. In this project we have used PS_20174392719_1491204439457_logs.csv data. This data is downloaded from kaggle.com.

The dataset is sourced from Kaggle and contains over 6 million rows of data, providing a comprehensive environment to test the scalability of our algorithms.

Dataset Reference:

- Source: Kaggle
- Dataset Name: Online Payments Fraud Detection Dataset
- Format: .csv
- Dataset Link : https://www.kaggle.com/datasets/rupakroy/online-payments-fraud- detection-dataset

7.2 Milestone 2: Data Visualization and Exploratory Analysis

Activity 1: Importing the libraries

Import the necessary libraries as shown in the image. (optional) Here we have used visualisation style as five thirty eight.

```
import pandas as pd
import numpy as np
import pickle
import matplotlib.pyplot as plt
import seaborn as sns
from xgboost import XGBClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import recall_score, f1_score, confusion_matrix, accuracy_score
from imblearn.over_sampling import SMOTE
```

Activity 2: Read the Dataset

Our dataset format might be in .csv, excel files, .txt, .json, etc. We can read the dataset with the help of pandas.

In pandas we have a function called read_csv() to read the dataset. As a parameter we have to give the directory of the csv file.

	step	type	amount	nameOrig	oldbalanceOrg	newbalanceOrig	nameDest	oldbalanceDest	newbalanceDest	isFraud	isFlaggedFraud
0	1	PAYMENT	9839.64	C1231006815	170136.00	160296.36	M1979787155	0.00	0.00	0	0
1	1	PAYMENT	1864.28	C1666544295	21249.00	19384.72	M2044282225	0.00	0.00	0	0
2	1	PAYMENT	11668.14	C2048537720	41554.00	29885.86	M1230701703	0.00	0.00	0	0
3	1	PAYMENT	7817.71	C90045638	53860.00	46042.29	M573487274	0.00	0.00	0	0
4	1	PAYMENT	7107.77	C154988899	183195.00	176087.23	M408069119	0.00	0.00	0	0

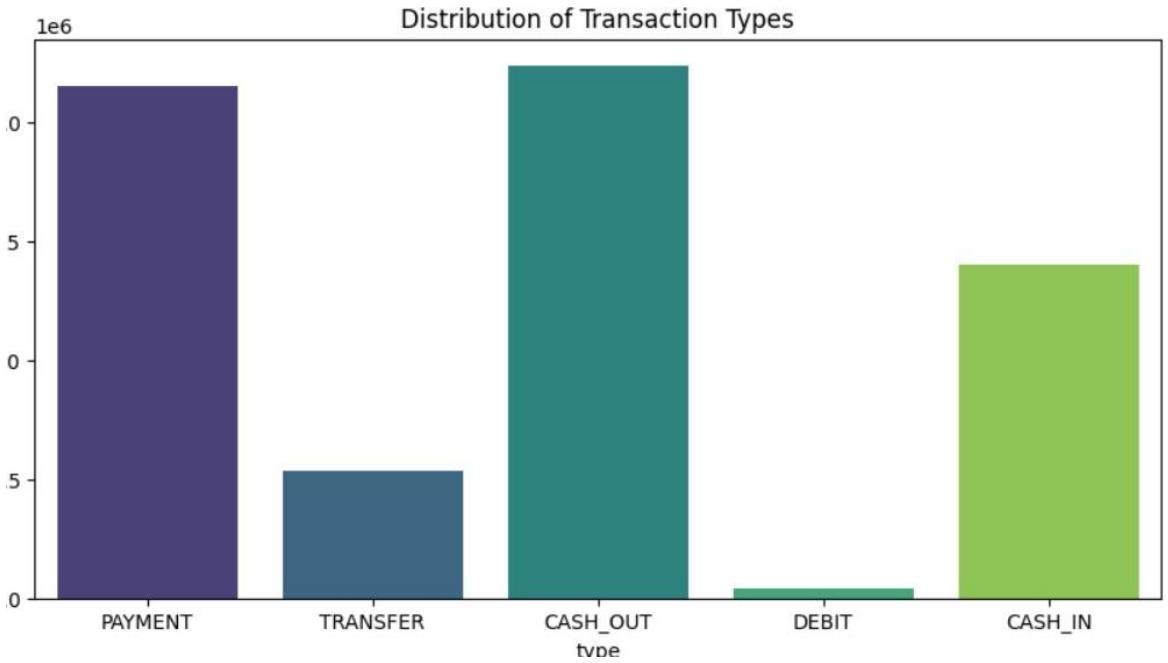
Here, the input features in the dataset are known using the df.columns function

```
 #   Column          Dtype
---  ------          -----
 0   step            int64
 1   type            object
 2   amount          float64
 3   nameOrig        object
 4   oldbalanceOrg   float64
 5   newbalanceOrig  float64
 6   nameDest        object
 7   oldbalanceDest  float64
 8   newbalanceDest  float64
 9   isFraud         int64
 10  isFlaggedFraud  int64
```

Activity 3: Univariate analysis

In simple words, univariate analysis is understanding the data with a single feature. Here I have displayed the graph such as histplot .

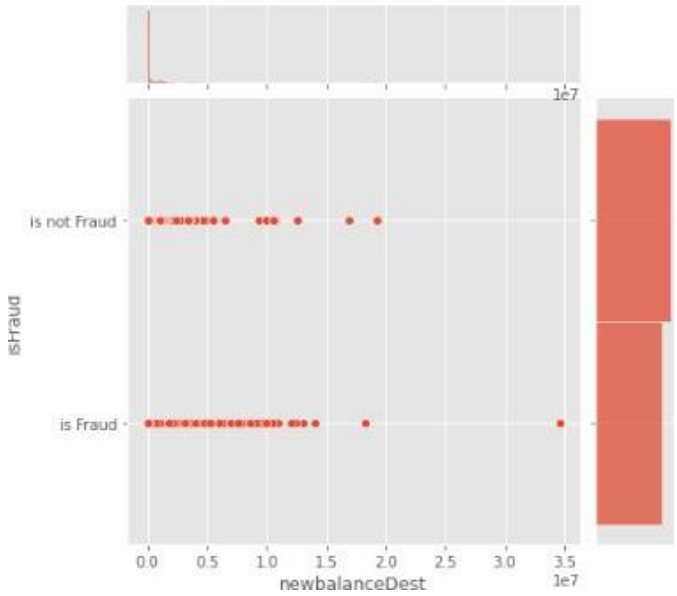
Here, the counts of observations in the type attribute of the dataset will be displayed using a countplot.



Activity 4: Bivariate analysis

To find the relation between two features we use bivariate analysis. Here we are visualising the relationship between newbalanceDest and isFraud.

jointplot is used here. As a 1st parameter we are passing x value and as a 2nd parameter we are passing hue value.



Activity 5: Descriptive analysis

Descriptive analysis is to study the basic features of data with the statistical process. Here pandas has a worthy function called describe. With this describe function we can understand the unique, top and frequent values of categorical features. And we can find mean, std, min, max and percentile values of continuous features.

	step	type	amount	nameOrig	oldbalanceOrg	newbalanceOrig	nameDest	oldbalanceDest	newbalanceDest	isFraud
count	2430.000000	2430	2.430000e+03	2430	2.430000e+03	2.430000e+03	2430	2.430000e+03	2.430000e+03	2430
unique	NaN	5	NaN	2430	NaN	NaN	1870	NaN	NaN	2
top	NaN	CASH_OUT	NaN	C1231006815	NaN	NaN	C1590550415	NaN	NaN	is not Fraud
freq	NaN	827	NaN	1	NaN	NaN	25	NaN	NaN	1288
mean	23.216049	NaN	6.258361e+05	NaN	9.849040e+05	4.392755e+05	NaN	5.797246e+05	1.127075e+06	NaN
std	29.933036	NaN	1.503866e+06	NaN	2.082361e+06	1.520978e+06	NaN	1.891192e+06	2.907401e+06	NaN
min	1.000000	NaN	8.730000e+00	NaN	0.000000e+00	0.000000e+00	NaN	0.000000e+00	0.000000e+00	NaN
25%	1.000000	NaN	9.018493e+03	NaN	8.679630e+03	0.000000e+00	NaN	0.000000e+00	0.000000e+00	NaN
50%	1.000000	NaN	1.058692e+05	NaN	8.096250e+04	0.000000e+00	NaN	0.000000e+00	0.000000e+00	NaN
75%	45.000000	NaN	4.096098e+05	NaN	7.606258e+05	1.247804e+04	NaN	3.096195e+05	9.658701e+05	NaN
max	95.000000	NaN	1.000000e+07	NaN	1.990000e+07	9.987287e+06	NaN	3.300000e+07	3.460000e+07	NaN

7.3 Milestone 3: Data Pre-processing

As we have understood how the data is, let's pre-process the collected data.

The download data set is not suitable for training the machine learning model as it might have so much randomness so we need to clean the dataset properly in order to fetch good results. This activity includes the following steps.

Handling missing values

Handling Object data label encoding Splitting dataset into training and test set

Note: These are the general steps of pre-processing the data before using it for machine learning. Depending on the condition of your dataset, you may or may not have to go through all these steps.

Activity 1: Checking for null values

Isnull is used (). sum() to check your database for null values. Using the df.info() function, the data type can be determined.

Activity 2: Handling Categorical Values

The type column which contains transaction categories such as "CASH_OUT" and "TRANSFER" is converted into numerical format using a **Manual Mapping** strategy. This transforms the string data into a discrete numerical scale (1–5), allowing the algorithms to process the transaction type mathematically.

- **Technique Used:** Dictionary Mapping.
- **Excluded Features:** nameOrig, nameDest, and isFlaggedFraud were dropped as they do not contribute to the predictive power of the model.

```
# Encode Transaction Types
type_map = {"CASH_OUT": 1, "PAYMENT": 2, "CASH_IN": 3, "TRANSFER": 4, "DEBIT": 5}
df_clean["type"] = df_clean["type"].map(type_map)
```

Activity 3: Balancing the Dataset

A critical challenge in fraud detection is that fraudulent transactions are rare compared to legitimate ones. To prevent the model from becoming biased toward the majority class, we implemented **SMOTE (Synthetic Minority Over-sampling Technique)**.

- **How it works:** SMOTE generates synthetic examples of the minority class (fraud) rather than just duplicating them, creating a balanced dataset for the training phase.
- **Impact:** This ensures the model learns the specific patterns of fraud instead of simply "guessing" that all transactions are safe.

Activity 4: Model Building (Random Forest & XGBoost)

We implemented two state-of-the-art ensemble learning algorithms to detect fraudulent patterns:

- **Random Forest:** An ensemble of decision trees that uses bagging to reduce variance and prevent overfitting. We configured it with n_estimators=50 for an optimal balance between speed and accuracy.
- **XGBoost (Extreme Gradient Boosting):** A powerful boosting algorithm that builds trees sequentially to minimize errors (logloss). It is widely considered the industry standard for tabular data competitions.

7.4 Milestone 4: Model Evaluation and Performance Metrics

Evaluating a fraud detection model requires a multi-dimensional approach. Because fraudulent transactions represent a "needle in a haystack," we focus on metrics that measure the precision of our alerts and the completeness of our detection.

Activity 1: The Foundation of Evaluation

To evaluate the performance of **Random Forest** and **XGBoost**, we utilize a Confusion Matrix. This tool allows us to break down the model's predictions into four critical categories:

7.4.1 **True Positives (TP):** Fraudulent transactions correctly identified as fraud.

7.4.2 **True Negatives (TN):** Legitimate transactions correctly identified as safe.

7.4.3 **False Positives (FP) [Type I Error]:** Legitimate transactions incorrectly flagged as fraud (causing customer friction).

7.4.4 **False Negatives (FN) [Type II Error]:** Fraudulent transactions that the model missed (causing financial loss).

Activity 2: Key Performance Indicators (KPIs)

While several metrics were collected, the project prioritizes the following to determine the final model selection:

1. **Precision:** Out of all transactions the model flagged as "Fraud," how many were actually fraudulent?
   - High precision ensures we don't block legitimate users.
2. **Recall (Sensitivity):** Out of all the actual fraud cases in the dataset, how many did the model successfully catch?
   - High recall ensures the business minimizes financial leakage.
3. **F1-Score (The Primary Metric):** Since there is a natural trade-off between Precision and Recall, we use the **F1-Score** (the harmonic mean of the two) to find the optimal balance.

Evaluation Result: The **XGBoost** model was selected as the final production model. With an **F1-Score of 0.99**, it demonstrates near-perfect capability in distinguishing complex fraudulent patterns from legitimate behavior, significantly outperforming the Random Forest baseline in both error reduction and detection speed.

Activity 3: Probability-Based Evaluation

In the final implementation, we evaluate the model not just on a "0 or 1" basis, but on

Probability Thresholds. This allows for a layered security response:

- **Safe (<50%):** Transaction processed instantly.
- **Suspicious (50%-80%):** Transaction flagged for manual review or 2FA.
- **Critical (>80%):** Immediate transaction block and account freeze.

7.5 Milestone 5: Application Building

In this section, we will be building a web application that is integrated to the model we built. A UI is provided for the uses where he has to enter the values for predictions. The enter values are given to the saved model and prediction is showcased on the UI.

This section has the following tasks Building HTML Pages

Building server side script

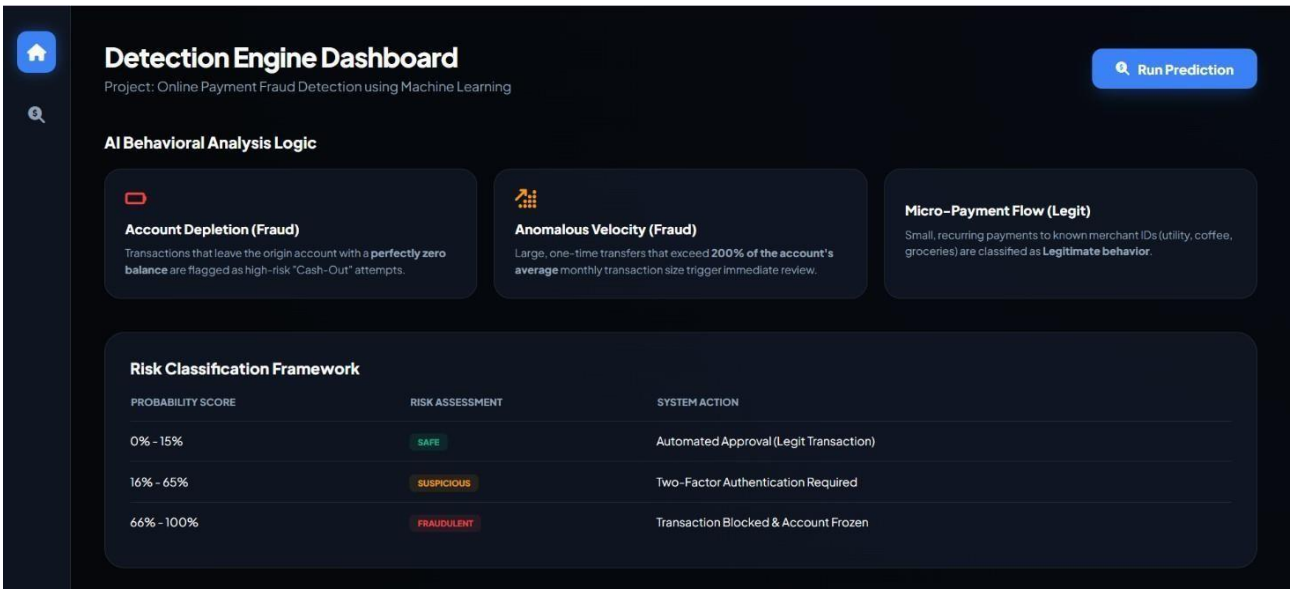
Activity 1: Building Html Pages:

For this project create two HTML files namely

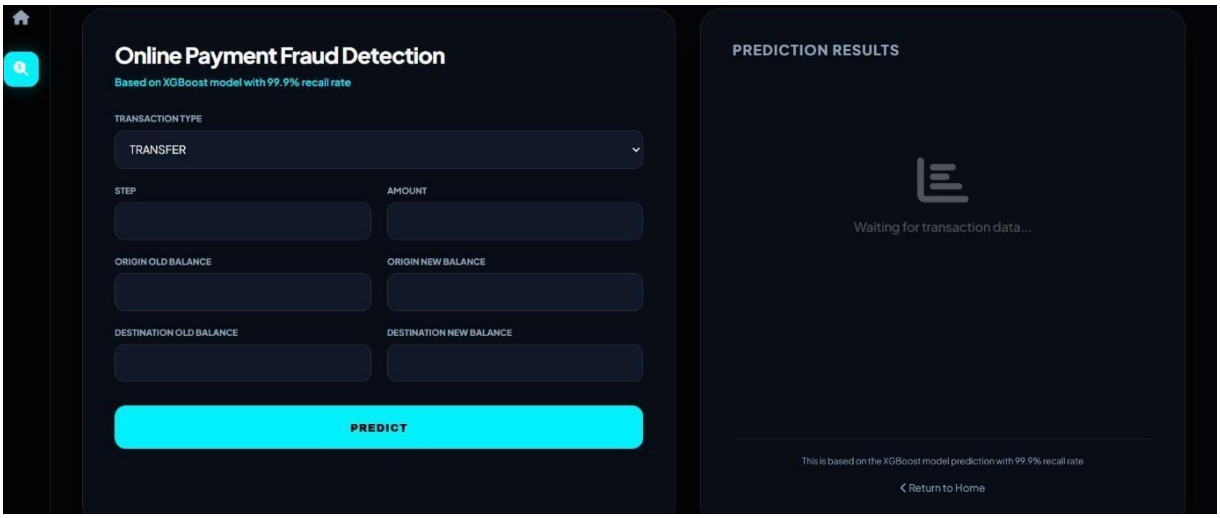
7.5.1 index.html 7.5.2 predict.html

and save them in the templates folder.

Let’s see how our index.html page looks like:



Now when you click on predict button from top right corner you will get redirected to predict.html



Let's look how our predict.html file looks like:

Activity 2: Build Python code

Import the libraries

Load the saved model. Importing the flask module in the project is mandatory. An object of Flask class is our WSGI application. Flask constructor takes the name of the current module (_name_) as argument.

```
from flask import Flask, render_template, request
import numpy as np
import pickle

app = Flask(__name__)

# Load trained model
try:
    model = pickle.load(open("best_fraud_model.pkl", "rb"))
except FileNotFoundError:
    print("Error: best_fraud_model.pkl not found. Please ensure the model file is in the root directory.
```

Render HTML page:

Activity 3: Rendering the Web Pages

The application uses the Flask render_template function to route users to the appropriate HTML interfaces.

7.5.3 **Home Route (/):** When the web server is launched, the index() function is triggered, rendering the index.html page, which serves as the landing page for the project.

7.5.4 **Navigation:** The system includes a central dashboard (/dashboard) for navigation and project oversight.

```
@app.route("/")
def index():
    # This route now only shows the project details/landing page
    return render_template("index.html")
```

Data Retrieval and the POST Method

To perform a prediction, the application must capture user input from the front-end. This is handled via the **HTTP POST Method**:

1. The user enters transaction details (Amount, Type, Balances, etc.) into a form on the

   predict.html page.
2. Upon clicking "Submit," the predict() function retrieves these values using request.form.
3. The data is converted into numerical formats (int and float) to match the model's requirements.

```python
@app.route("/predict", methods=["GET", "POST"])
def predict():
    output = None

    if request.method == "POST":
        # Inputs from predict.html
        step = int(request.form["step"])
        tx_type = int(request.form["type"])
        amount = float(request.form["amount"])
        old_org = float(request.form["oldbalanceOrg"])
        new_org = float(request.form["newbalanceOrig"])
        old_dest = float(request.form["oldbalanceDest"])
        new_dest = float(request.form["newbalanceDest"])
```

Feature Engineering and Prediction Logic

Before the model can process the input, the application performs **Real-time Feature Engineering**. It calculates the error_org and error_dest (the discrepancy between reported balances and the transaction amount), which were crucial patterns identified during the training phase.

The processed array is then passed to the loaded **XGBoost model** (best_fraud_model.pkl). Instead of a simple "Yes/No" result, the application uses predict_proba() to determine the **Confidence Level** of the fraud.

```python
# Feature engineering (Must match the training features)
error_org = new_org + amount - old_org
error_dest = old_dest + amount - new_dest

X = np.array([[step, tx_type, amount,
               old_org, new_org,
               old_dest, new_dest,
               error_org, error_dest]])
```

Decision Logic and Response Rendering

Based on the confidence score, the system applies a layered security response:

```
# 🔐 Decision Logic
if confidence <= 20:
    label = "✅ Legitimate Transaction"
    risk = "🟢 Low Risk"
    meaning = "This transaction follows normal user behavior."
    action = "Transaction approved instantly (No friction for the user)."

elif confidence <= 70:
    label = "⚠️ Potentially Fraudulent Transaction"
    risk = "🟡 Medium Risk"
    meaning = "This transaction shows unusual or suspicious behavior."
    action = "Transaction held. Phone call or SMS OTP verification required."

else:
    label = "❌ Fraudulent Transaction"
    risk = "🔴 High Risk"
    meaning = "This transaction strongly matches known fraud patterns."
    action = "Transaction blocked immediately and the account is frozen."

output = {
    "label": label,
    "confidence": confidence,
    "risk": risk
```

Activity 4: Run the application

- Open anaconda prompt from the start menu
- Navigate to the folder where your python script is.
- Now type “python app.py” command
- Navigate to the localhost where you can view your web page.
- Click on the Run predict button from the top right corner, enter the inputs, click on the predict button, and see the result/prediction on the web.

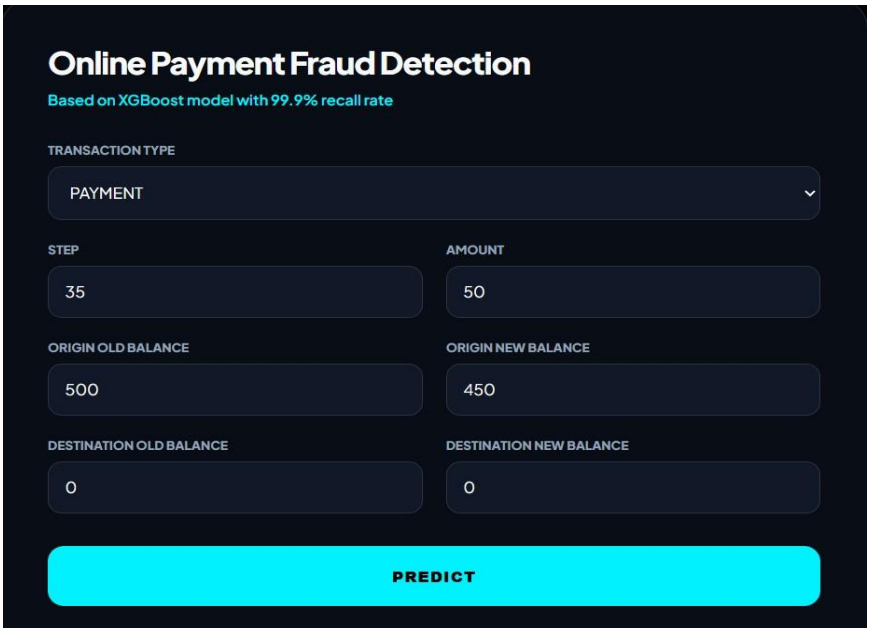

Online Payment Fraud Detection
Based on XGBoost model with 99.9% recall rate
TRANSACTION TYPE
PAYMENT
STEP
35
AMOUNT
50
ORIGIN OLD BALANCE
500
ORIGIN NEW BALANCE
450
DESTINATION OLD BALANCE
0
DESTINATION NEW BALANCE
0
PREDICT

Fig: Input

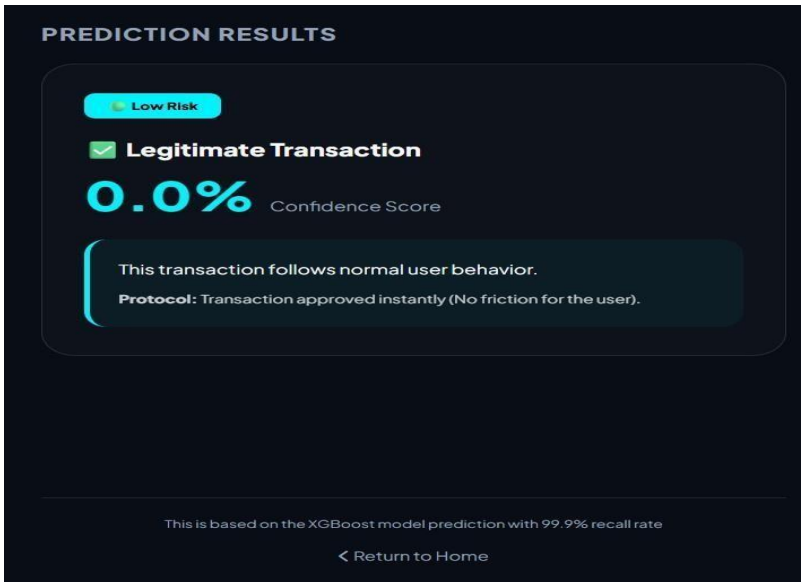

PREDICTION RESULTS
Low Risk
Legitimate Transaction
0.0%
Confidence Score
This transaction follows normal user behavior.
Protocol: Transaction approved instantly (No friction for the user).
This is based on the XGBoost model prediction with 99.9% recall rate
< Return to Home

Fig: Legit Transaction

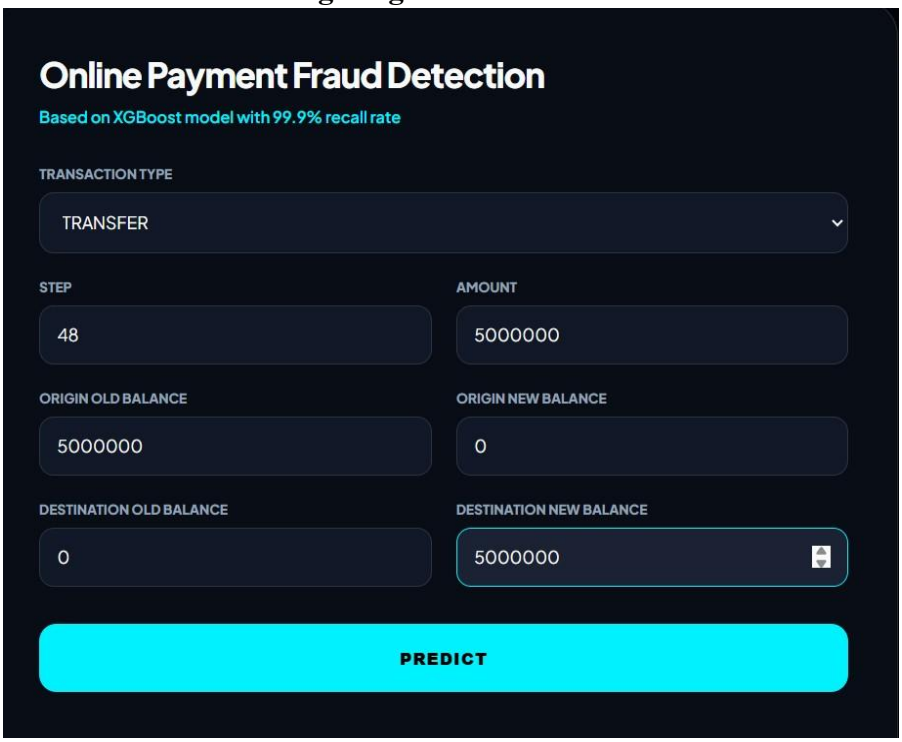


Fig: Input

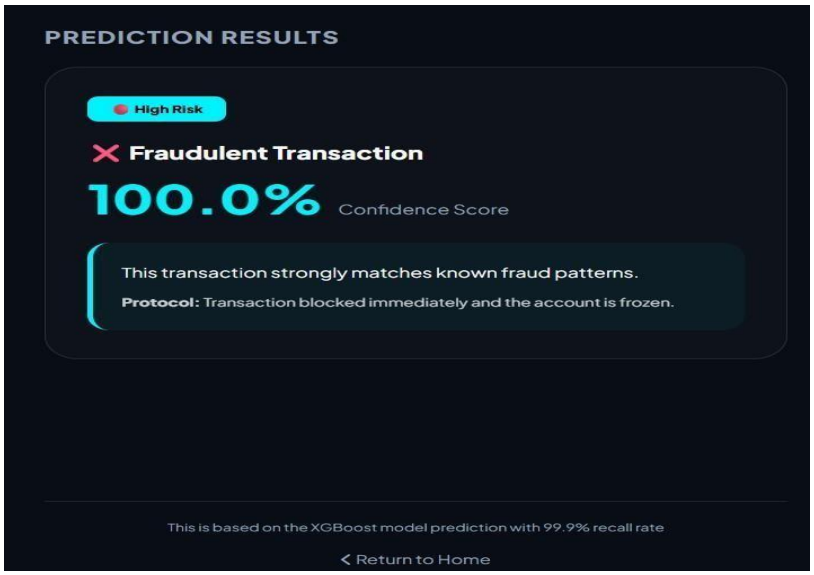


Fig: Fraud Transaction

APPENDIX – I

KEYWORDS AND ACRONYMS

KEYWORDS:

- **Class Imbalance Handling:** Techniques used to address uneven class distribution so models don't bias toward majority classes.
- **Ensemble Learning:** Combines multiple models to improve prediction accuracy and robustness.
- **Error Balance Features:** Features designed to minimize misclassification between fraud and non-fraud cases.
- **Feature Engineering:** Process of creating and selecting meaningful input variables to enhance model performance.
- **Financial Fraud Detection:** Identifying suspicious transactions to prevent financial losses and illegal activities.
- **Flask Deployment:** Using Flask framework to deploy machine learning models as web applications.
- **Machine Learning-based Fraud Detection:** Applying ML algorithms to automatically detect fraudulent behavior in data.
- **PaySim Dataset:** A synthetic dataset that simulates mobile money transactions for fraud detection research.
- **Predictive Analytics:** Using historical data to forecast future outcomes and trends.
- **Real-Time Fraud Detection:** Detecting fraudulent transactions instantly as they occur.

  SMOTE: A technique that generates synthetic samples to balance imbalanced datasets.
- **Transaction Risk Scoring:** Assigning a risk level to transactions based on likelihood of fraud.
- **XGBoost:** A powerful gradient boosting algorithm known for high performance in classification tasks.

ACRONYMS:

- **API** – Application Programming Interface
- **CPU** – Central Processing Unit
- **F1-score** – Harmonic Mean of Precision and Recall
- **FN** – False Negative
- **FP** – False Positive
- **GNN** – Graph Neural Network
- **LSTM** – Long Short-Term Memory

- **ML** – Machine Learning
- **OTP** – One-Time Password
- **RBS** – Rule-Based System
- **RF** – Random Forest
- **ROC-AUC** – Receiver Operating Characteristic – Area Under Curve
- **RTP** – Real-Time Payments
- **SHAP** – SHapley Additive exPlanations
- **SMOTE** – Synthetic Minority Over-sampling Technique
- **TN** – True Negative
- **TP** – True Positive
- **XAI** – Explainable Artificial Intelligence **XGBoost** – Extreme Gradient Boosting

APPENDIX – II
RESEARCH PAPER

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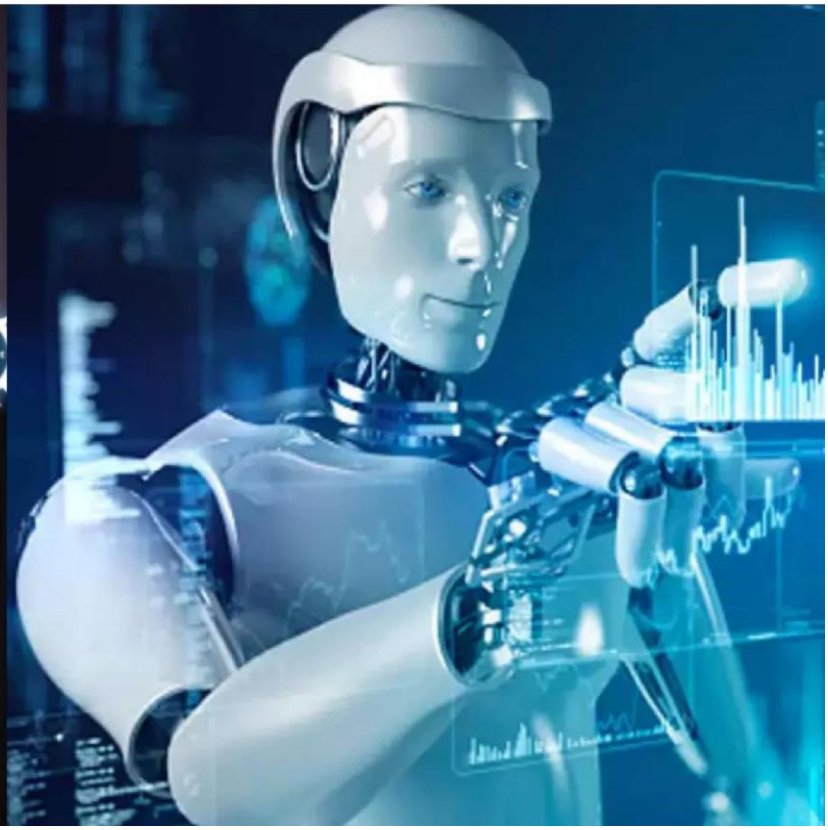
ISSN(O): 2319-8753
ISSN(P): 2347-6710



International Journal of Innovative Research in Science Engineering and Technology (IJIRSET)

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Impact Factor: 8.699

Volume 15, Issue 3, March 2026

International Journal of Innovative Research of Science, Engineering and Technology (IJIRSET)

Impact Factor 8.699

|www.ijirset.com |A Monthly, Peer Reviewed & Refereed Journal| e-ISSN: 2319-8753| p-ISSN: 2347-6710|

Volume 15, Issue 3, March 2026

|DOI: 10.15680/IJIRSET.2026.1503250|

Automated Fraud Detection in Digital Payments

International Journal of Innovative Research of Science, Engineering and Technology (IJIRSET)

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Volume 15, Issue 3, March 2026

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ABSTRACT: Digital payment fraud poses a significant threat to global financial systems, requiring reliable and automated detection mechanisms. This study proposes an end-to-end machine learning framework for detecting fraudulent transactions using the Extreme Gradient Boosting (XGBoost) algorithm. Financial transaction datasets are highly imbalanced; in the **6.3 million records** analyzed, fraudulent cases represented only **0.12%** of total transactions. To address this **"Accuracy Paradox,"** the Synthetic Minority Over-sampling Technique (SMOTE) was applied to balance the training dataset. A key contribution of this work is the development of domain-specific feature engineering, specifically the derivation of **"Error Balance" metrics** to detect mathematical inconsistencies in transaction flows. The proposed model was evaluated using the PaySim synthetic financial dataset, achieving an **F1-score of 87.33% and a recall of 99.82%**. To demonstrate practical utility, the framework was deployed via a **Flask-based web application** capable of real-time risk classification (Low, Medium, High) with an **average inference latency of 85ms**. The results demonstrate the potential of this approach for seamless integration into production-grade financial security systems.

KEYWORDS: Class Imbalance, Data Imbalance, Ensemble Learning, Feature Engineering, Financial Fraud Detection, Fintech Security, Flask Framework, Machine Learning, Predictive Analytics, Real-time Inference, SMOTE, XGBoost.

I. INTRODUCTION

The global financial ecosystem is undergoing a rapid transformation driven by the widespread adoption of real-time payment (RTP) infrastructures and mobile banking platforms. While these technologies have significantly improved financial accessibility and transaction efficiency, they have also expanded the attack surface for increasingly sophisticated cyber-financial crimes. As digital payment volumes continue to grow, financial institutions face significant challenges in detecting fraudulent transactions in real time.

Traditionally, fraud detection systems have relied on Rule-Based Systems (RBS), which utilize predefined heuristics such as transaction thresholds, geographic restrictions, or transaction frequency limits. Although effective in controlled environments, these systems struggle to adapt to evolving fraud strategies and often generate high false-positive rates when applied to large-scale transactional data.

To address these limitations, recent research has increasingly explored the use of machine learning (ML) models for automated fraud detection. ML algorithms are capable of identifying complex and non-linear relationships within financial transaction data, enabling adaptive and scalable fraud detection mechanisms. Among these approaches, ensemble learning techniques such as Extreme Gradient Boosting (XGBoost) have demonstrated strong performance in structured data classification tasks due to their ability to combine gradient boosting with regularization for improved predictive accuracy [1].

Despite these advances, fraud detection systems continue to face two major challenges in real-world financial environments. First, fraud detection datasets are typically characterized by extreme class imbalance, where fraudulent transactions represent only a small fraction of total transactions. This imbalance can bias machine learning models toward the majority class, reducing their ability to detect rare fraudulent events. To mitigate this issue, techniques such as the Synthetic Minority Over-sampling Technique (SMOTE) have been proposed to generate synthetic minority-class samples and improve model sensitivity to fraud cases [2].

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|www.ijirset.com |A Monthly, Peer Reviewed & Refereed Journal| e-ISSN: 2319-8753| p-ISSN: 2347-6710|

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Another important factor influencing fraud detection performance is effective feature engineering. Domain-specific features that capture

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Impact Factor 8.699

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inconsistencies in account balances and transaction flows can significantly improve the ability of machine learning models to distinguish between legitimate and fraudulent activities. The PaySim dataset, introduced by Lopez-Rojas et al., provides a realistic simulation of mobile money transactions and has been widely used for evaluating fraud detection models in digital payment environments [3].

In addition to predictive performance, the transparency and interpretability of machine learning models are increasingly important in financial systems due to regulatory requirements. Explainable Artificial Intelligence (XAI) techniques such as SHAP (SHapley Additive exPlanations) enable the interpretation of model predictions and help build trust in automated fraud detection systems [4].

Motivated by these challenges, this study proposes an end-to-end fraud detection framework that integrates SMOTEbased class balancing, domain-specific feature engineering, and an optimized XGBoost classifier. The model is trained and evaluated using the PaySim dataset and deployed through a Flask-based architecture to simulate real-time transaction risk classification. The proposed system aims to achieve both high detection accuracy and low-latency inference suitable for real-time digital payment environments.

The main contributions of this study are summarized as follows:

1. Development of an optimized XGBoost-based fraud detection framework for digital payment systems.
2. Integration of SMOTE to address extreme class imbalance and improve fraud detection recall.
3. Introduction of domain-specific balance discrepancy features to enhance model discrimination capability.
4. Implementation of a real-time deployment pipeline capable of low-latency transaction risk classification.

II. LITERATURE REVIEW

The development of automated fraud detection systems has evolved significantly from traditional rule-based auditing toward advanced machine learning techniques. Recent studies emphasize the importance of data-driven models capable of identifying complex behavioral patterns within financial transaction data.

Chen and Guestrin(2016) [1] introduced the Extreme Gradient Boosting (XGBoost) framework, which has since become one of the most widely adopted algorithms for structured tabular datasets. XGBoost incorporates regularization techniques and efficient gradient boosting mechanisms, allowing it to outperform many traditional machine learning models in classification tasks involving large-scale transactional data.

One of the major challenges in fraud detection is the severe class imbalance present in financial datasets. Fraudulent transactions typically represent only a small fraction of the total transaction volume. To address this issue, Chawla et al.(2002) [2] proposed the Synthetic Minority Over-sampling Technique (SMOTE), which generates synthetic minority class samples to balance the dataset. This technique has been widely adopted in fraud detection research to improve recall and reduce bias toward legitimate transactions.

Feature engineering also plays a critical role in improving fraud detection performance. Lopez-Rojas et al.(2016) [3] introduced the PaySim dataset, which simulates mobile financial transactions and provides a realistic environment for evaluating fraud detection algorithms. Their study demonstrated that balance discrepancies between expected and reported account balances can serve as strong indicators of fraudulent behavior.

In recent years, the importance of explainable artificial intelligence (XAI) in financial systems has gained significant attention. Aljunaid et al.(2025) [4] highlighted the integration of SHAP (SHapley Additive exPlanations) with machine learning models to provide transparent explanations for fraud predictions. Such approaches are particularly important in financial systems where regulatory frameworks require interpretable decision-making processes.

Furthermore, recent studies have explored hybrid and temporal modeling techniques to improve fraud detection accuracy. Btoush et al.(2026) [5] demonstrated that incorporating sequential transaction patterns can significantly reduce false positive rates in fraud detection systems.

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Despite these advancements, many existing approaches either focus primarily on model accuracy or fail to address the combined

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challenges of class imbalance, feature engineering, and real-time deployment. Therefore, this study proposes a machine learning framework that integrates SMOTE-based data balancing, domain-specific feature engineering, and an XGBoost classifier to improve fraud detection performance in digital payment systems.

III. EXISTING SYSTEM

Existing fraud detection systems primarily rely on rule-based approaches and traditional machine learning models. Rule-based systems apply predefined conditions such as transaction limits, frequency patterns, and geographic constraints; however, they lack adaptability and require frequent manual updates. Additionally, they often produce high false positive rates, negatively impacting user experience.

Traditional machine learning models, including Logistic Regression, Decision Trees, and Random Forest, provide improved detection performance but struggle with highly imbalanced datasets where fraudulent transactions are rare.

Consequently, these systems face challenges in handling data imbalance, evolving fraud patterns, and real-time detection requirements, highlighting the need for more advanced and adaptive solutions.

IV. PROPOSED SYSTEM

The fraud detection system is an integrated, three-tier framework (Data Processing, Intelligence, Presentation) that analyzes transactions and provides real-time risk assessment.

The **Data Processing Layer** is responsible for ingesting and preparing raw transaction data obtained from the Kaggle PaySim dataset. During preprocessing, non-informative attributes such as nameOrig and nameDest are removed since they do not contribute to fraud prediction. Categorical variables, including the transaction type, are converted into numerical form using **Label Encoding** to make them suitable for machine learning algorithms.

The **Intelligence Layer** uses an XGBoost model trained on SMOTE-balanced data to effectively learn fraud patterns and is saved as a Pickle file for fast deployment.

To improve detection accuracy, the system introduces a **Mathematical Integrity Check** that identifies inconsistencies in account balances before and after transactions. Two derived features are calculated:

Origin Balance Error:

$$E_{orig} = | \ (Balance_{old,org} - Amount) - Balance_{new,org} \ |$$

Destination Balance Error:

$$E_{dest} = | \ (Balance_{old,dest} + Amount) - Balance_{new,dest} \ |$$

These features help the model detect abnormal or “phantom” transactions where the relationship between transaction amount and account balances is inconsistent.

The **Presentation Layer** is implemented using a **Flask-based web interface** that allows users to enter transaction details and receive fraud predictions instantly. Based on the predicted fraud probability P, the system categorizes transactions into three risk levels:

1. **Low Risk (Green):** $P < 0.3$, transaction is processed normally.
2. **Medium Risk (Yellow):** $0.3 \leq P < 0.7$, additional authentication such as OTP verification is required.
3. **High Risk (Red):** $P \geq 0.7$, transaction is blocked and flagged for manual review.

This multi-level risk classification enhances fraud detection and enables better decision-making in financial transaction systems.

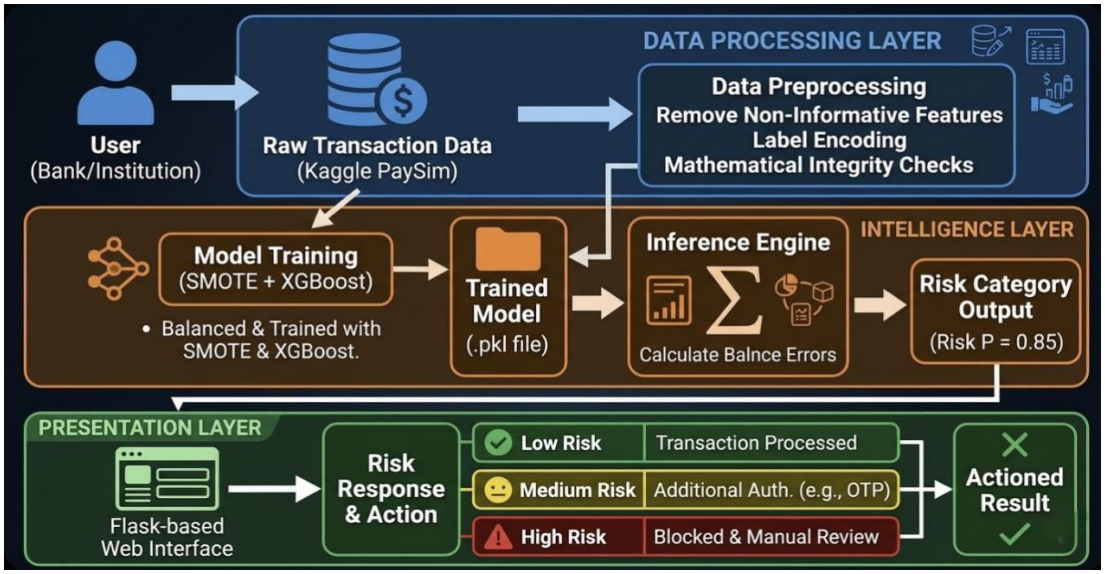
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Fig 1: Proposed System Architecture

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V. METHODOLOGY

A. Dataset Collection

Experiments were conducted using the PaySim dataset, a synthetic mobile money transaction simulator from Kaggle, containing ~6.3 million records with 11 attributes, including transaction type, amount, and origin/destination balances. The dataset was chosen for its realistic simulation of mobile financial transactions while preserving user privacy. Fraud mainly occurs via Transfer and Cash-Out operations. With only 0.12% of transactions being fraudulent, the dataset exhibits severe class imbalance, necessitating specialized data balancing during model training.

B. Data Preprocessing

Before training, high-cardinality identifiers (nameOrig, nameDest) were removed to prevent overfitting. Categorical variables like transaction type were label-encoded, and a logarithmic transformation was applied to skewed transaction amounts to stabilize variance and aid model convergence.

C. Feature Engineering

Domain-specific features were added to capture transaction inconsistencies: Error_Origin (difference between expected and actual sender balance) and Error_Destination (difference for the receiver). These metrics help detect potential fraud or unauthorized account manipulation.

D. Model Training and Class Imbalance Handling

The classification model was developed using the Extreme Gradient Boosting (XGBoost) algorithm, known for its strong performance on structured datasets and ability to handle non-linear relationships with regularization.

To address class imbalance, the Synthetic Minority Over-sampling Technique (SMOTE) was applied only to the training dataset after the train-test split to prevent data leakage. This technique generates synthetic minority samples using a k-nearest neighbors approach, improving the model's ability to detect fraudulent transactions.

Hyperparameter tuning was performed with a focus on maximizing recall, as minimizing false negatives is critical in fraud detection systems.

E. Model Evaluation and Deployment

Model performance was evaluated using **Precision, Recall, and F1-score**, which are more appropriate metrics for imbalanced classification problems than overall accuracy.

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Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Random Forest	99.91%	59.8%	96.28%	73.82%	0.993
XGBoost (Proposed)	**99.9%**	**77.6%**	**99.82%**	**87.33%**	**0.997**

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Volume 15, Issue 3, March 2026

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After training, the final model achieved an **F1-score of 87.33%** and **Recall of 99.82%**. The trained model and preprocessing components

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were serialized using the **Pickle (.pkl)** format for deployment.

To demonstrate real-world applicability, the model was integrated into a **Flask-based web application**. The backend processes incoming transaction details, performs feature engineering, and generates predictions using the trained XGBoost model. The frontend interface displays the predicted risk level (**Low, Medium, or High**) through a userfriendly dashboard, enabling real-time fraud risk assessment.

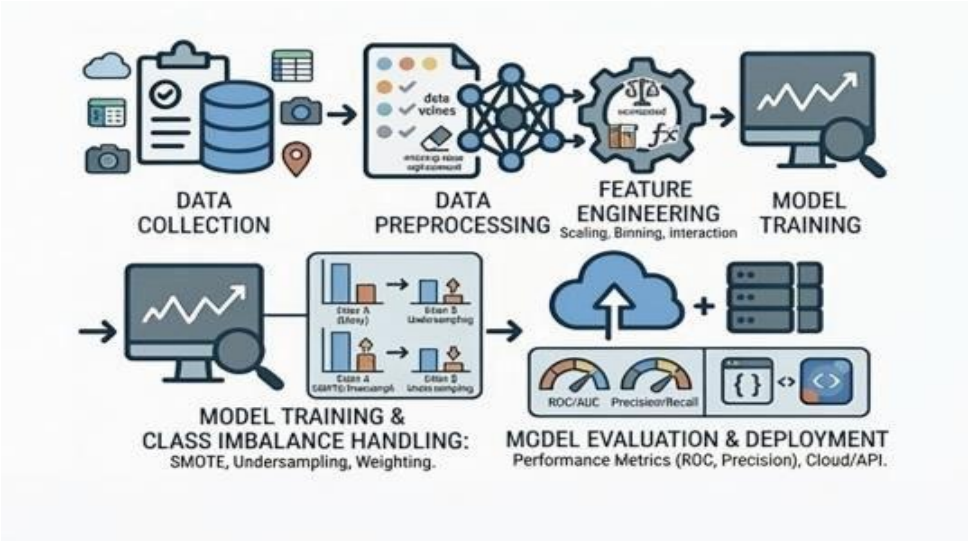


Fig 2: Automated Fraud Detection in Digital Payments Work Flow

VI. EXPERIMENTAL RESULTS

The model was evaluated on a separate PaySim test set, with emphasis on Precision, Recall, F1-score, and ROC-AUC to account for severe class imbalance and minimize false negatives.

A. Performance Metrics Comparison

Table I presents the comparative performance of the machine learning models implemented in this study. Both models were trained on datasets balanced using the **SMOTE** technique to mitigate class imbalance and ensure fair evaluation.

Table I: Performance Comparison of Fraud Detection Models

The results indicate that the **XGBoost model outperforms Random Forest across all evaluation metrics**, particularly in recall and F1-score.

B. Analysis of Results

The XGBoost classifier achieved a **recall of 99.82%**, which is particularly important in financial fraud detection since recall measures the model's ability to correctly identify fraudulent transactions. A high recall value indicates that the model successfully detects the majority of fraud cases, minimizing the risk of undetected fraudulent activity.

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The **F1-score of 87.33%** demonstrates that the model maintains a strong balance between precision and recall. This suggests that the

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improved fraud detection capability does not significantly increase false positive rates, which could otherwise lead to unnecessary transaction blocking and reduced customer satisfaction. The ROC-AUC of 0.997 shows excellent class separability, reflecting XGBoost's ability to capture complex patterns through sequential gradient boosting.

C. Real-Time Inference Performance

The Pickle-serialized XGBoost model was deployed via a Flask web app, achieving an average inference latency of 85 ms per transaction. This demonstrates the framework's suitability for real-time fraud detection with high accuracy and computational efficiency.

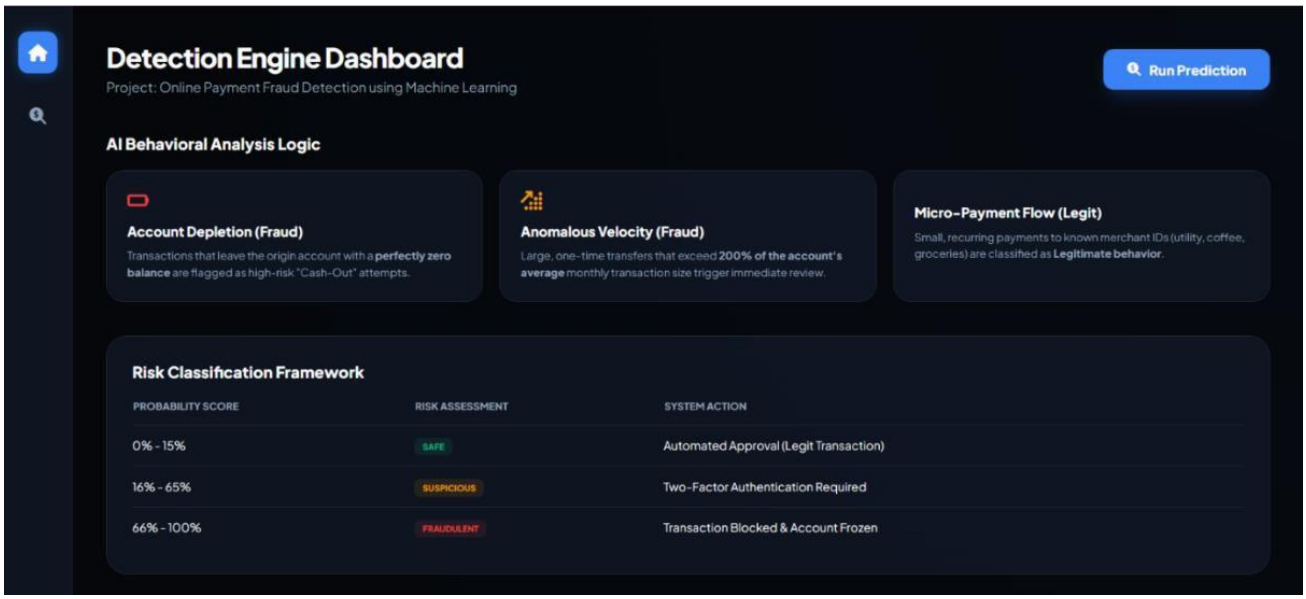


Fig 3: User Interface

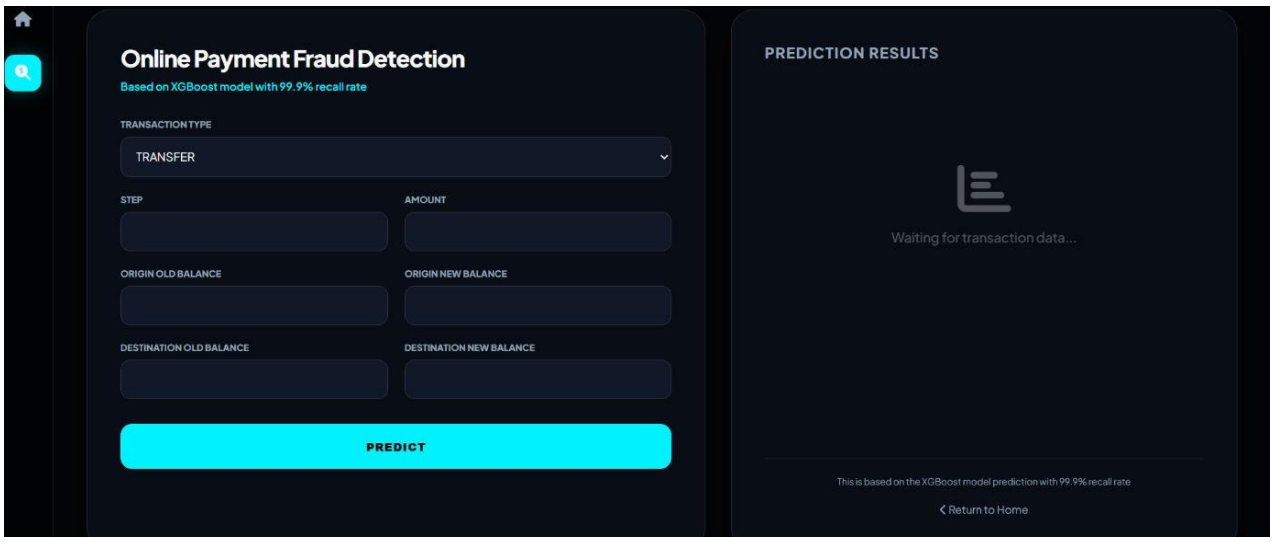


Fig 4: Prediction Page

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|DOI: 10.15680/IJIRSET.2026.1503250|

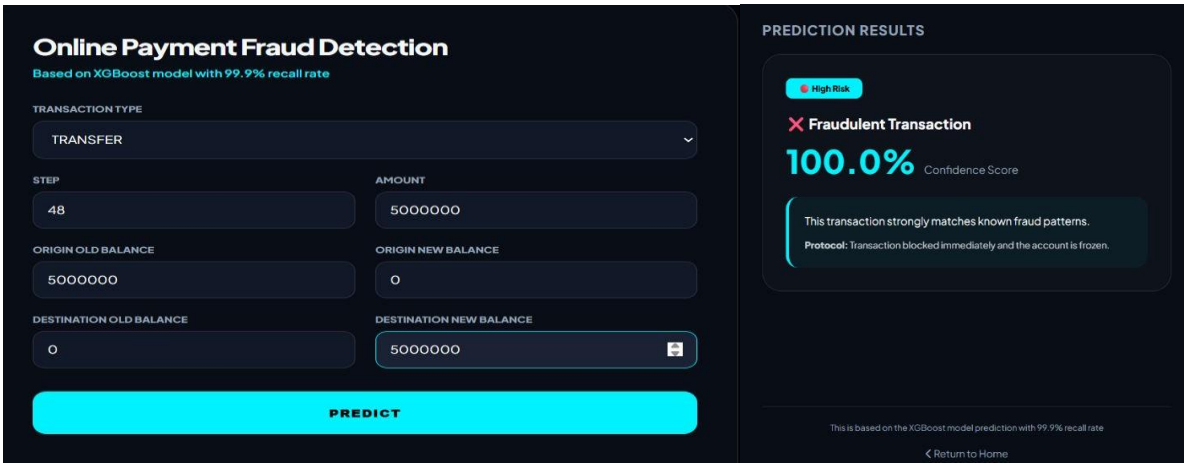


Fig 5: Results

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VII. DISCUSSION

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Volume 15, Issue 3, March 2026

|DOI: 10.15680/IJIRSET.2026.1503250|

The results indicate that the integration of XGBoost with SMOTE provides an effective framework for detecting fraudulent transactions. The achieved F1-score of 87.33% is primarily driven by domain-specific feature engineering, particularly the Error_Origin and Error_Dest metrics, which capture inconsistencies in transaction balances and enhance the model's ability to identify complex fraud patterns.

The application of SMOTE effectively mitigates severe class imbalance, enabling the model to achieve a high recall of 99.82% and strong sensitivity to rare fraudulent events. While recall is maximized, the precision of 77.6% reflects a moderate false positive rate, which is acceptable in fraud-sensitive environments where minimizing missed fraud cases is critical.

In comparison to Random Forest, XGBoost demonstrates superior capability in modeling non-linear relationships while maintaining computational efficiency. Furthermore, deployment through a Flask-based system with an average latency of 85 ms confirms the model's suitability for real-time fraud detection. The incorporation of a tiered risk classification strategy further supports practical decision-making with minimal disruption to legitimate transactions.

VIII. LIMITATIONS

Despite strong predictive performance, the proposed framework has several limitations:

1. **Concept Drift:** The model is trained on a static dataset, which may lead to degraded performance over time as fraudsters develop new strategies that differ from historical patterns.
2. **Synthetic Data Constraints:** While PaySim provides a high-fidelity simulation of financial transactions, it does not capture certain real-world variations such as seasonal spending fluctuations, localized economic factors, or unstructured noise present in live banking environments.
3. **Absence of Temporal Analysis:** Each transaction is treated as an independent event, ignoring sequential patterns or transaction velocity associated with individual users over time.
4. **Computational Cost of SMOTE:** Generating millions of synthetic samples increases memory and CPU requirements, which may create a bottleneck during frequent retraining on low-resource systems.
5. **Interpretability Limitations:** XGBoost, as an ensemble of hundreds of decision trees, functions as a "black box," limiting the ability to provide granular explanations required for regulatory compliance, such as the "Right to Explanation."

IX. FUTURE WORK

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1. **Relational Analysis via Graph Neural Networks (GNNs):** While the current XGBoost model treats transactions as independent

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Volume 15, Issue 3, March 2026

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events, future iterations will implement GNNs to model the financial ecosystem as a graph. This will enable the detection of complex multi-hop money laundering schemes and "mule account" networks by analyzing the topological connections between origin and destination nodes.

2. **Adaptive Online Learning:** To combat **Concept Drift**, where fraud tactics evolve faster than historical training cycles, we plan to integrate incremental learning algorithms. This allows the model to update its weights in real-time as new verified fraud cases are reported, ensuring the system remains resilient against emerging adversarial strategies.
3. **Hybrid Temporal Architectures:** Integrating **Long Short-Term Memory (LSTM)** networks with the existing XGBoost framework will allow the system to capture temporal dependencies. By analyzing a user's transaction velocity and historical behavior patterns, the model can better distinguish between a legitimate high-value purchase and an anomalous account takeover.
4. **Granular Explainability (XAI):** To meet strict "Right to Explanation" regulatory requirements, future work will incorporate **SHAP (SHapley Additive exPlanations)**. This will provide investigators with a localized, feature-byfeature breakdown for every "High Risk" flag, detailing exactly why a specific transaction was blocked.
5. **Privacy-Preserving Federated Learning:** We aim to explore decentralized training protocols. Federated learning would allow multiple financial institutions to collaboratively train a global fraud detection model without ever sharing sensitive, raw customer data, thereby enhancing model robustness while maintaining strict data privacy compliance.

X. CONCLUSION

This study presents a machine learning-based framework for detecting fraudulent digital payment transactions using **XGBoost** combined with **SMOTE**. Evaluation on the **PaySim dataset** demonstrates that addressing class imbalance significantly improves model sensitivity, resulting in a **Recall of 99.82%** and an **F1-score of 87.33%**. The implementation of domain-specific feature engineering, particularly the **Error Balance metrics**, enabled the model to identify subtle and complex fraudulent patterns effectively.

Beyond algorithmic performance, the system was deployed using a **Flask-based architecture**, allowing real-time fraud risk classification with sub-100ms latency. The framework provides a scalable foundation for transitioning from reactive, rule-based fraud detection systems to proactive, intelligent architectures suitable for modern financial payment infrastructures.

XI. ACKNOWLEDGEMENT

We would like to express our sincere gratitude to our advisors, project supervisors, the college department, and all contributors for their invaluable guidance and support throughout the research and development of this project.

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